JACC: CLINICAL ELECTROPHYSIOLOGY VOL. 8, NO. 5, 2022

PUBLISHED BY ELSEVIER

STATE-OF-THE-ART REVIEW

Congenital Long QT Syndrome

Andrew D. Krahn, MD,[a] Zachary Laksman, MD, MSc,[a] Raymond W. Sy, MBBS, PhD,[b] Pieter G. Postema, MD, PhD,[c] Michael J. Ackerman, MD, PhD,[d,e,f] Arthur A.M. Wilde, MD, PhD,[c,g] Hui-Chen Han, MBBS, PhD[a,h]

ABSTRACT

Congenital long QT syndrome (LQTS) encompasses a group of heritable conditions that are associated with cardiac repolarization dysfunction. Since its initial description in 1957, our understanding of LQTS has increased dramatically. The prevalence of LQTS is estimated to be ∼1:2,000, with a slight female predominance. The diagnosis of LQTS is based on clinical, electrocardiogram, and genetic factors. Risk stratification of patients with LQTS aims to identify those who are at increased risk of cardiac arrest or sudden cardiac death. Factors including age, sex, QTc interval, and genetic background all contribute to current risk stratification paradigms. The management of LQTS involves conservative measures such as the avoidance of QT-prolonging drugs, pharmacologic measures with nonselective β-blockers, and interventional approaches such as device therapy or left cardiac sympathetic denervation. In general, most forms of exercise are considered safe in adequately treated patients, and implantable cardioverter-defibrillator therapy is reserved for those at the highest risk. This review summarizes our current understanding of LQTS and provides clinicians with a practical approach to diagnosis and management. (J Am Coll Cardiol EP 2022;8:687-706) © 2022 by the American College of Cardiology Foundation.

Congenital long QT syndrome (LQTS) encompasses a group of heritable conditions that are associated with cardiac repolarization dysfunction.[1-4] The initial description of a hereditary syndrome causing QT interval prolongation, recurrent syncope, congenital deafness, and sudden cardiac death (SCD) was made by Jervell and Lange-Nielsen[5] in 1957. Early work in LQTS sought to identify patients based on the clinical phenotype of prolonged QT on the surface electrocardiogram (ECG).[6,7] With the advent of genetic testing, more patients are now identified on the basis of a pathogenic variant with varying degrees of penetrance and phenotypic expression.[8,9] Due to the potential risk for SCD and the availability of effective treatment options, it is vital for clinicians to be able to accurately identify and manage patients suspected of having LQTS. In this review, we summarize the current understanding of LQTS and provide a practical framework for its diagnosis, risk stratification, and treatment with a focus on the 3 major autosomal dominant LQTS genotypes: LQT1, LQT2, and LQT3 (**Central Illustration, Table 1**).

From the [a]Center for Cardiovascular Innovation, Heart Rhythm Services, Division of Cardiology, University of British Columbia, Vancouver, BC, Canada; [b]Faculty of Medicine and Health, The University of Sydney, Sydney, New South Wales, Australia; [c]Department of Clinical and Experimental Cardiology, Heart Center, Amsterdam University Medical Centers, Amsterdam, the Netherlands; [d]Department of Cardiovascular Medicine, Division of Heart Rhythm Services, Windland Smith Rice Genetic Heart Rhythm Clinic, Mayo Clinic, Rochester, Minnesota, USA; [e]Department of Pediatric and Adolescent Medicine, Division of Pediatric Cardiology, Mayo Clinic, Rochester, Minnesota, USA; [f]Departments of Molecular Pharmacology and Experimental Therapeutics, Windland Smith Rice Sudden Death Genomics Laboratory, Mayo Clinic, Rochester, Minnesota, USA; [g]European Reference Network for Rare and Low Prevalence Complex Diseases of the Heart (ERN GUARD-Heart), Academic University Medical Center, Amsterdam, the Netherlands; and the [h]Victorian Heart Institute, Monash University, Clayton, VIC, Australia.

Kenneth Ellenbogen, MD, served as Guest Associate Editor for this paper.

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

Manuscript received August 27, 2021; revised manuscript received February 16, 2022, accepted February 16, 2022.

ISSN 2405-500X/$36.00 https://doi.org/10.1016/j.jacep.2022.02.017

ABBREVIATIONS AND ACRONYMS

APD = action potential duration

CE = cardiac events

ECG = electrocardiogram

ICD = implantable cardioverter-defibrillator

JLN = Jervell and Lange-Nielsen

LCSD = left cardiac sympathetic denervation

LQTS = long QT syndrome

SAE = serious arrhythmic events

SCD = sudden cardiac death

TdP = torsade de pointes

PATHOPHYSIOLOGY

POTASSIUM CHANNEL STRUCTURE AND FUNCTION. Voltage-gated potassium channels are transmembrane proteins that are essential for the regulation of cardiac repolarization, and are composed of an α-subunit and β-subunits.[10,11] The α-subunit is composed of 4 homologous domains—each consisting of 6 segments—that coassemble into a pore-forming channel (Figure 1), allowing the movement of potassium ions across the plasma membrane along their electrochemical gradient.[12,13] The auxiliary β-subunits are regulatory proteins (eg, cytoplasmic proteins, single transmembrane spanning domains, and transport related proteins), that modulate the function of the α-subunit through cell surface expression and gating.[10,11]

Two subtypes of voltage-gated potassium channels—delayed rectifier channels $K_V7.1$ and $K_V11.1$, accounting for I_{Ks} (slow) and I_{Kr} (rapid), respectively—are primarily responsible for the outward potassium current during phase 3 of the ventricular action potential (Figure 2),[14] which is critical for cardiac repolarization.[12] Under normal circumstances, the physiological function of both $K_V7.1$ and $K_V11.1$ are potentiated by adrenergic stimulation.[15,16]

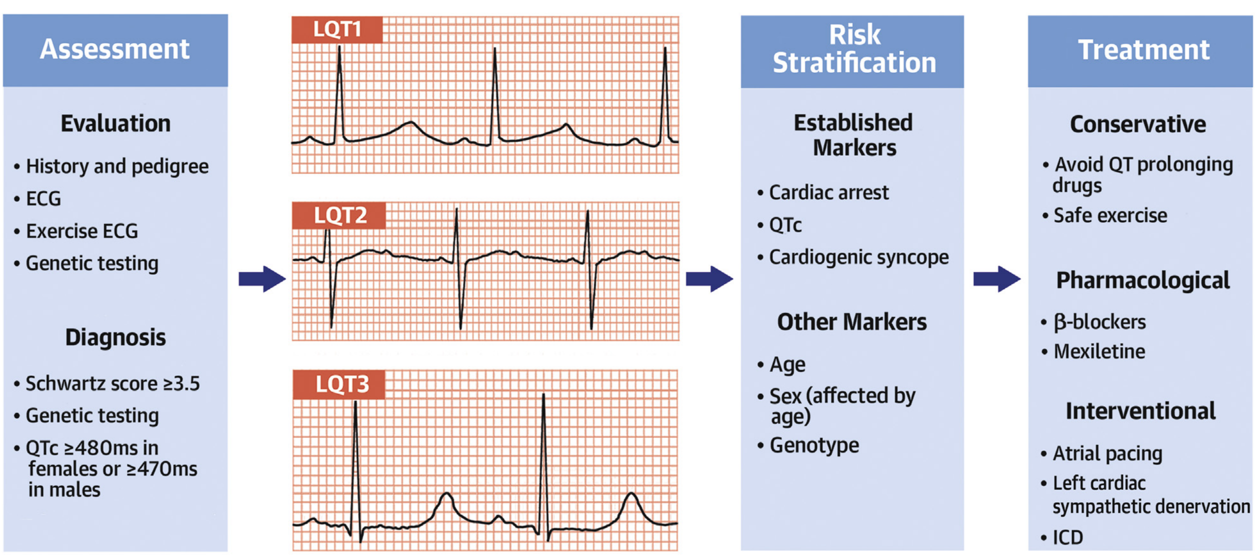
HIGHLIGHTS

- Recently, there have been significant advances in our understanding of congenital LQTS.
- This review provides practical recommendations for the diagnosis and treatment of congenital LQTS.
- Further research is required regarding the pathophysiological understanding and risk stratification in congenital LQTS.

SODIUM CHANNEL STRUCTURE AND FUNCTION. Voltage-gated sodium channels are similar in structure to voltage-gated potassium channels, with the predominant isoform being $Na_V1.5$ in the human heart.[17] Cardiac depolarization occurs as a result of sodium channel activation whereby a conformational change in the α-subunit leads to the rapid influx of sodium ions (I_{Na}).[18]

GENETICS OF LQTS. The majority of LQTS cases are caused by loss-of-function variants in voltage-gated potassium channels.[3] Variants in genes encoding for

CENTRAL ILLUSTRATION Clinical Approach to Congenital Long QT Syndrome



Krahn AD, et al. J Am Coll Cardiol EP. 2022;8(5):687-706.

ECG = electrocardiogram; ICD = implantable cardioverter-defibrillator; LCSD = left cardiac sympathetic denervation.

TABLE 1 Diagnosis and Management Summary for LQTS

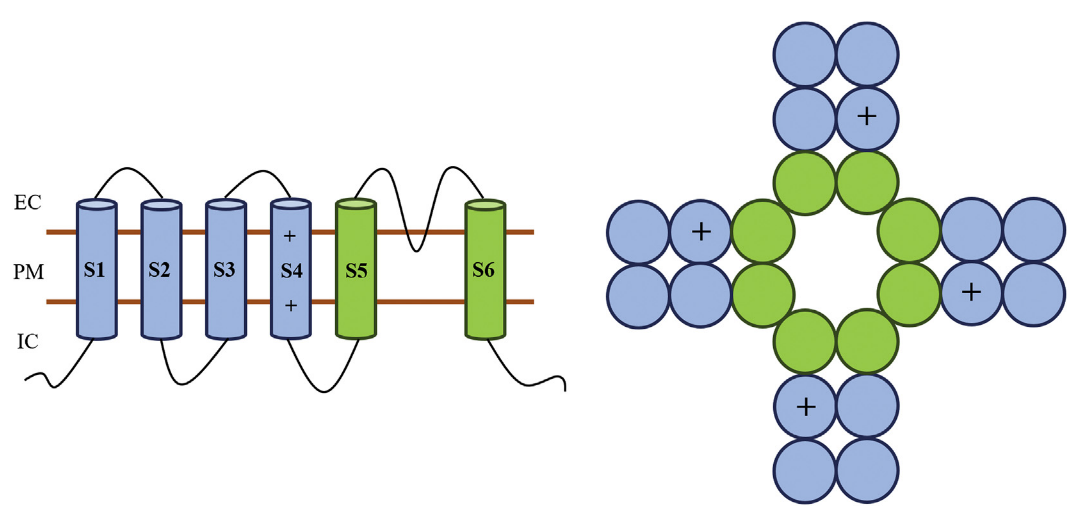
Diagnosis		
At Risk	**Evaluation and Testing**	**Diagnostic Criteria**
Symptomatic • Cardiogenic syncope • Ventricular arrhythmias • Resuscitated cardiac arrest *Asymptomatic* • Prolonged QTc interval • Abnormal T wave • Family screening	*Initial* • Clinical—syncope, family history, medical history, medications • ECG—QTc interval, T-wave morphology • Exercise test—QTc interval at 4-min recovery • Echocardiogram—exclude structural abnormalities *Discretionary* • Holter monitor • Further cardiac imaging as indicated • Genetic testing with involvement of genetics expert	*Definite* • Schwartz-score ≥3.5 • Pathogenic or likely pathogenic variant in LQTS gene • QTc interval ≥500 ms on repeated ECG[a] *Probable* • QTc interval between 480 and 499 ms on repeated ECG[a] *Possible* • LQTS risk score 2.0-3.5 in the presence of a family history of LQTS
Management		
Conservative	**Pharmacologic**	**Interventional**
Avoid QT prolonging drugs and triggers • For all patients with definite LQTS • Recommended for all patients with probable LQTS *Re-evaluation* • Consider re-evaluating patients with possible LQTS after 1-2 y	*β-blockers* • QTc interval ≥470 ms in patients with definite LQTS • Patients with cardiogenic syncope and definite LQTS • Patients with probable LQTS • Consider in patients with possible LQTS • Nonselective agents preferred *Mexiletine* • Consider as adjunctive therapy in LQT3 • Consider as adjunctive therapy in LQT2 with QTc interval ≥500 ms or syncope on β-blocker	*ICD* • Secondary prevention in resuscitated cardiac arrest • Consider for primary prevention in high-risk patients—breakthrough syncope or QTc interval ≥500 ms despite β-blocker (± mexiletine and LCSD), JLN syndrome *Atrial overdrive pacing* • During acute ventricular arrhythmias *LCSD* • β-Blocker intolerance • Arrhythmic events despite β-blockers ± ICD

[a]In the absence of medications or disorders known to affect these electrocardiographic findings.

ECG = electrocardiogram; ICD = implantable cardioverter-defibrillator; JLN = Jervell and Lange-Nielsen; LCSD = left cardiac sympathetic denervation; LQTS = long QT syndrome.

$K_V7.1$ and $K_V11.1$ account for ~80% of all genotype-positive LQTS cases.[3] LQT1 is caused by pathogenic variation in *KCNQ1*, which encodes for the α-subunit of $K_V7.1$,[3,10] whereas LQT2 is related to pathogenic variation in *KCNH2*, which encodes for the α-subunit of $K_V11.1$.[3,10] These variants cause a reduction of outward potassium current during phase 3 of the ventricular action potential (Figure 2).[14] Variants in

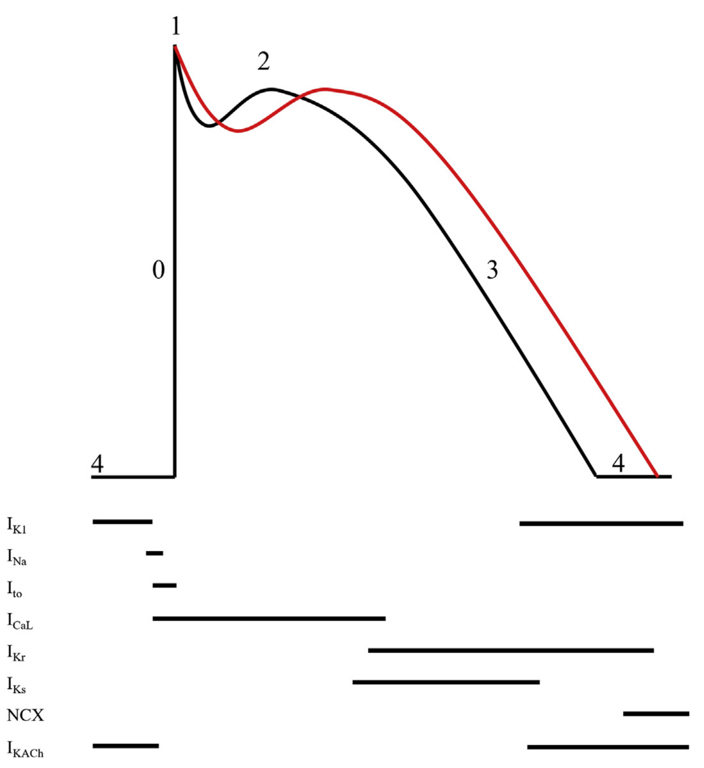
FIGURE 1 Schematic Representation of Voltage-Gated Potassium Channel



The α-subunit is composed of a cytoplasmic N-terminus, 4 homologous domains (D_I-D_{IV}) each composed of 6 transmembrane proteins (S1-S6), interdomain links, extracellular P-loops (between S5 and S6), and a cytoplasmic C-terminus. S4 is a voltage sensing domain, whereas S5 and S6 are ion pore forming domains. **Left image** depict one α-subunit domain. **Right image** depicts 4 homologous domains that coassemble into a pore-forming channel. EC = extracellular; IC = intracellular; PM = plasma membrane.

JACC: CLINICAL ELECTROPHYSIOLOGY VOL. 8, NO. 5, 2022
MAY 2022:687-706

FIGURE 2 **Action Potential**



The **black line** indicates normal ventricular action potential; the **red line** indicates increased action potential duration in long QT syndrome. Action potential phases: 0, rapid depolarization; 1, rapid/early repolarization; 2, plateau; 3, terminal repolarization; 4, resting potential.

genes encoding for the β-subunits of $K_v7.1$ and $K_v11.1$ have also been reported in patients With LQTS (*KCNE1* causing LQT5 and *KCNE2* causing LQT6, respectively).[19] Recent studies suggest a modulatory role that is associated with a weaker clinical phenotype,[20-22] although compound heterozygous variants in *KCNE1* result in the autosomal recessive Jervell and Lange-Nielsen (JLN) syndrome.[23]

Additionally, variants affecting other cardiac ion channels are also described in LQTS. LQT3 is caused by a gain-of-function variant in the *SCN5A* gene, encoding for $Na_V1.5$,[3,10] which causes persistent I_{Na} during the plateau phase of the action potential, with resultant prolongation of QT duration caused by delayed repolarization.[24]

ACTION POTENTIAL DURATION AND ARRHYTHMOGENESIS. The net effect of the ion channel disturbances in LQTS is prolongation of the action potential duration (APD) through alterations in ion channel activation and inactivation, or ion channel expression. Using induced pluripotent stem cells from 2 patients with LQT1 resulting from a missense variation in the *KCNQ1* gene, Moretti et al[25] showed that the ventricular myocyte APD at 90% repolarization was 554 ms in LQT1 compared with 373 ms in control subjects, which occurred as a result of a 70% to 80% reduction in I_{Ks}. Similar findings of APD prolongation have been demonstrated in other forms of LQTS.[26-28]

Under normal circumstances, sympathetic input with β-adrenergic stimulation of cardiomyocytes results in pleomorphic effects on various ion channels, including $K_V7.1$, $K_V11.1$, and $Na_V1.5$, resulting in the overall augmentation of depolarization and repolarization kinetics.[29] For example, I_{Ks} is enhanced by adrenergic stimulation, but this is diminished in the setting of defective $K_v7.1$ channels in patients with LQT1.[29] In patients with LQT2, β-adrenergic stimulation results in an increase of depolarizing currents but reduced repolarizing currents caused by defective $K_v11.1$ channels.[29] In patients with LQT3, there is an increase in late I_{Na}.[29] The resultant effect of these imbalances between depolarization and repolarization is lengthening of the APD,[30] with the occurrence of early afterdepolarizations, which can then trigger torsade de pointes (TdP).[25,26,31]

EPIDEMIOLOGY

The prevalence of LQTS is estimated to be ~1:2,000 based on a large newborn evaluation cohort, with a slight female predominance.[9,32] However, there may be considerable overlap in the baseline QTc interval between those with genotype-positive LQTS compared with genotype-negative family members,[9,33,34] whereas 1% to 10% of the general population may have a QTc interval ≥450 ms.[34,35] Conversely, up to 40% of those with genotype-positive LQTS may have a baseline QTc interval that is within the normal range.[8,9,36-38] LQTS accounts for 5% to 10% of young SCD cases referred for autopsy (diagnosed through molecular genetic testing and/or phenotype testing of family members),[39] and ~15% of cases with “unexplained” resuscitated cardiac arrest (after exclusion of coronary artery disease and structural heart disease).[40,41]

DIAGNOSIS

GUIDELINE RECOMMENDATIONS. The diagnosis of LQTS remains anchored to the LQTS diagnostic criteria score (also known as the ‘Schwartz-score’), which is an important tool for the initial clinical

TABLE 2 Schwartz Score

		Points
ECG findings[a]		
A	QTc interval[b], ms	
	≥480	3
	460-479	2
	450-459 (males)	1
B	QTc interval[b] ≥480 ms at 4-min recovery from exercise stress test	1
C	Torsade de pointes[c]	2
D	T-wave alternans	1
E	Notched T wave in 3 leads	1
F	Low heart rate for age[d]	0.5
Clinical history		
A	Syncope[c]	
	With stress	2
	Without stress	1
B	Congenital deafness	0.5
Family history[e]		
A	Family member with definite LQTS	1
B	Unexplained sudden cardiac death age <30 y among immediate family members	0.5

Score ≤1, low probability; 1.5 to 3, intermediate probability; ≥3.5, high probability. Reproduced with permission from Schwartz et al.[3] [a]In the absence of medications or disorders known to affect these electrocardiographic findings. [b]QTc interval calculated by Bazett formula QTc = QT/√(RR). [c]Mutually exclusive. [d]Resting HR <2nd percentile for age. [e]Same family member cannot be used for both A and B.

Abbreviations as in Table 1.

assessment of patients suspected of having LQTS (Table 2).[3] It is composed of ECG findings, clinical history, and family history. Importantly, ECG findings should be interpreted in the absence of QT-prolonging agents or other reversible causes. A Schwartz-score of ≥3.5 carries a specificity of 99% (with sensitivity between 19% and 36%) for the diagnosis of LQTS.[37,42]

Expert consensus guidelines from the Heart Rhythm Society, European Heart Rhythm Association, and Asia Pacific Heart Rhythm Society indicates that LQTS is definitely diagnosed in the setting of a Schwartz-score ≥3.5, the presence of a pathogenic variant or repeated measures of QTc interval ≥500 ms in the absence of QT-prolonging drugs,[42] although we would recommend involvement of a genetics expert if diagnosing LQTS primarily based on genetic results.

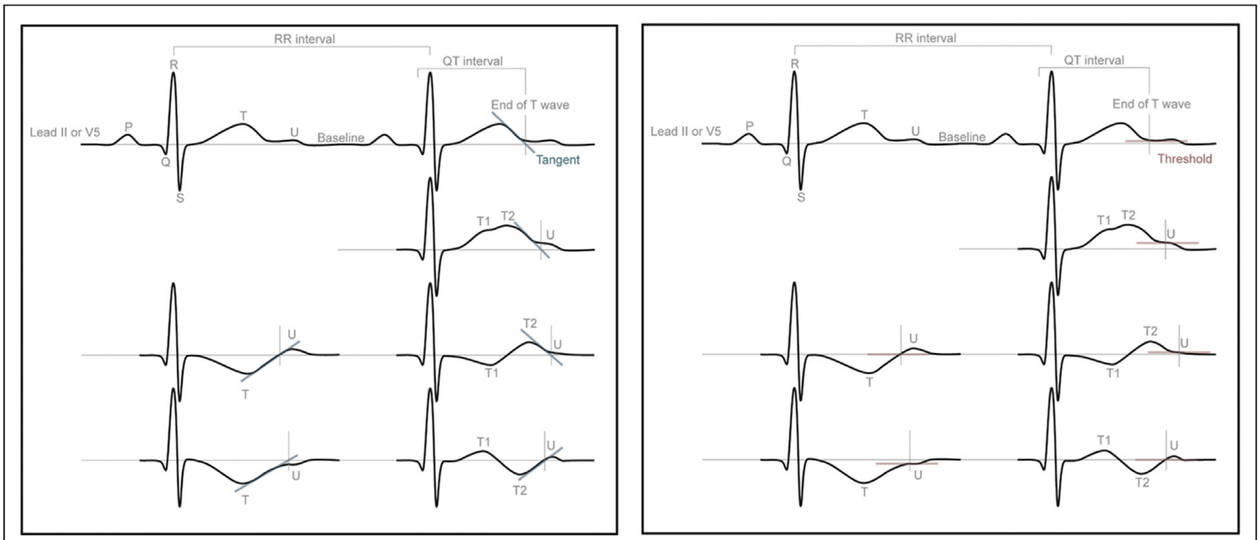
An important consideration is that patients who are diagnosed with "acquired" LQTS may in fact have underlying congenital LQTS that is unmasked with QT-prolonging drugs.[43] In an international cohort of patients with apparent acquired LQTS, about one-fourth of patients harbored a pathogenic LQTS gene variant.[44] Hence, patients with apparent acquired LQTS should be evaluated after drug cessation.

CLINICAL MANIFESTATIONS. Recurrent syncope, caused by TdP,[45] is frequently described in patients with LQTS. Classically, these episodes are triggered by exercise (particularly swimming) in patients with LQT1, and by sudden auditory stimuli (such as alarm clocks) in LQT2.[46-48] Additionally, episodes may also occur in the setting of emotional stress (for LQT1, LQT2, and LQT3),[48] during sleep or rest, but without arousal (particularly for LQT3),[48] or in the postpartum period for female patients with LQT2.[49] In patients who present with syncope, a detailed clinical assessment is required to differentiate likely cardiogenic syncope from other potential causes such as vasovagal syncope.[50]

Other clinical manifestations in patients with LQTS include seizures, atrial arrhythmias, and SCD or resuscitated cardiac arrest. There is also appreciable overlap between LQTS and epilepsy.[51] This is especially true in patients with LQT2 who harbor specific *KCNH2* variants,[51-53] and it is recognized that disturbance of potassium homeostasis in the hippocampus may result in epileptic activity.[51] Perturbations of cardiac repolarization seen in LQTS may also affect the atria.[54] Consequently, there is an increased risk of atrial arrhythmias, although this appears most evident for patients under the age of 60 with LQT3.[55] As noted, LQTS may also be identified after an episode of resuscitated cardiac arrest or at postmortem after SCD, supporting the development of population screening programs.[48,56] Finally, with the advent of genetic testing, a significant proportion of patients are now asymptomatic at the time of clinical diagnosis.[42,57] This shift in diagnostic paradigm is in contrast to the original seminal research from the International Long QT Registry that enrolled a cohort of patients and family members with a relatively severe disease phenotype.[58]

BASELINE ECG. The surface ECG remains fundamental in the initial assessment for LQTS. In general, calculation of the QT interval in lead II or V_5 offers the best combination of measurability, identification of gene variant carriers, and identification of high-risk individuals.[59] Suitable alternative leads include V_2 or V_3, which may provide a longer measured QT interval,[60] but is associated with lower predictive power in the identification of gene variant carriers compared with control subjects.[59] The QT interval is measured from the beginning of the QRS complex to the end of the T wave. Standardized assessment using either a tangent or threshold technique (Figure 3) enables high interobserver and intraobserver reproducibility and diagnostic accuracy.[9,61] Although both techniques are valid and accurate, associated cutoff

FIGURE 3 **Tangent and Threshold Methods for QT Measurement**



Tangent technique: the end of the T-wave is the point where the tangent to the steepest point of the terminal limb of the T-wave intersects with the isoelectric baseline.	**Threshold technique:** the end of the T-wave is the intersection of the terminal limb of the T-wave with the isoelectric baseline; when interrupted by a U-wave, it was defined as the nadir between the T- and U-wave.

Reproduced with permission from Vink et al. [9]

values for distinguishing LQTS patients from control subjects vary, whereby the tangent method results in QT measurements that are ~10 ms shorter.[9] Clinicians should be careful in distinguishing the second component of a biphasic T-wave from a U-wave, and exclude any present U-waves in the measurement of the QT interval.[9] An online tool that provides an estimate about the likelihood of a patient having LQTS is available at the University of Amsterdam Medical Center's QT calculator.

Given the heart rate dependence of the QT interval,[62] the rate-corrected QTc interval is used for the diagnosis of LQTS. Various formulas exist for QT correction,[63] with subsequent effects for the diagnostic cutoffs for LQTS[9]; readers are directed to work from Vandenberk et al[63] and Vink et al[9] for a detailed comparison of these techniques. However, the formula that is most commonly implemented in clinical practice was developed by Bazett in 1920,[62] defined as: $QTc = QT/\sqrt{(RR)}$. Although it overestimates the QTc interval compared with other methods,[63,64] the Bazett formula is applied in the current diagnostic criteria for LQTS.[3] The ECG is a contextual test, and thus pretest probability based on clinical circumstances is crucial. In a healthy population, most slightly prolonged QTc intervals represent the healthy upper limit of normal findings, and not genetically mediated LQTS. A resting QTc interval ≥450 ms in men or ≥460 ms in women—in the absence of medications or other conditions affecting the QTc interval—increases the pretest possibility for LQTS.[3] As noted however, these cutoff values may be exceeded in 1% to 10% of the general population,[34] whereas the QTc interval may be within the normal range in up to 40% of genotype-positive patients.[9,37] Furthermore, an additional 30% may have a borderline QTc interval (up to 470 ms in males and up to 480 ms in females).[8] Hence, additional clinical evaluation is critical before assigning a diagnosis of LQTS.

In addition to prolongation of the QTc interval, patients with LQTS may exhibit morphological T-wave changes. The initial descriptions for T-wave changes in LQTS included broad, biphasic, notched, and the presence of beat-to-beat T-wave alternans.[65,66] Subsequent genotype studies identified typical T-wave patterns for the 3 most common forms of LQTS (Figure 4): LQT1 patients frequently have a

FIGURE 4 **Characteristic T-Wave Changes in LQTS**

LQTS = long QT syndrome.

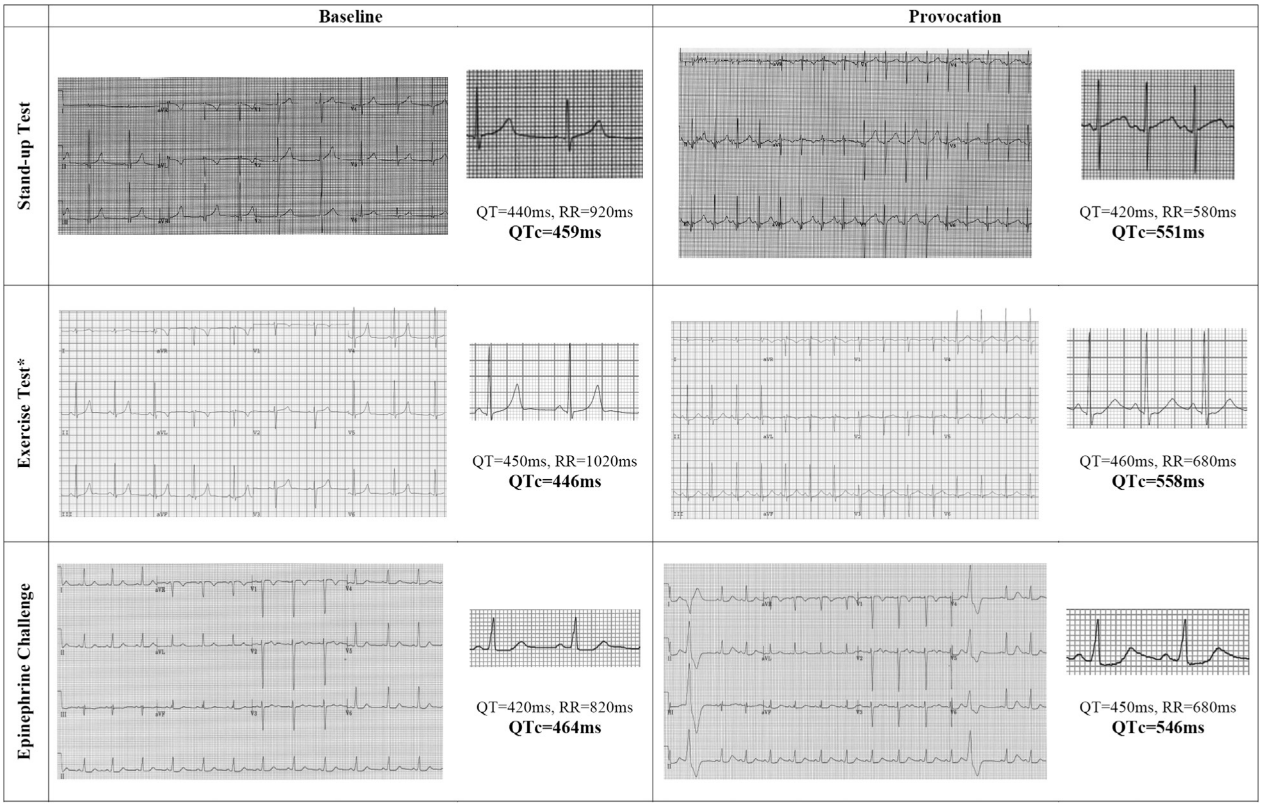
broad-based T-wave pattern; LQT2 patients frequently have a low-amplitude bifid T-wave pattern; and LQT3 patients frequently have a peaked or biphasic T-wave pattern that can be late onset or asymmetrical.[2,67,68] Interestingly, the presence of giant T waves (sometimes termed T-U waves due to their size) are reported to herald episodes of TdP.[69]

PROVOCATION TESTING. In cases of diagnostic uncertainty, provocation testing is often helpful for the identification of gene variant carriers.[70] Provocation testing for LQTS involves assessment of repolarization in response to autonomic changes,[71] particularly during the acceleration and deceleration of heart rate. Although the QT and QTc interval are known to be influenced by heart rate alterations in normal individuals,[62] a maladaptive repolarization response is frequently seen in patients with LQTS.

Exercise testing in LQTS allows for an extended assessment of QT and T-wave changes during both acceleration and deceleration phases.[8,72,73] Outcomes have been described for various forms of exercise testing, including the standard or modified Bruce protocol treadmill test and bicycle ergometry.

JACC: CLINICAL ELECTROPHYSIOLOGY VOL. 8, NO. 5, 2022
MAY 2022:687-706

FIGURE 5 **Changes in QTc Interval With Provocation Testing**



*Provocation electrocardiogram performed at 4-minute recovery.

A protocol is described for QT measurements irrespective of the mode of exercise (Supplemental Table 1).[8] Although various measures of repolarization have been investigated, the measure with the highest diagnostic value during exercise testing is the QTc interval at 4-minute recovery (**Figure 5**, Supplemental Table 2),[8,70] with a QTc interval ≥445 ms offering 90% sensitivity and 90% specificity for diagnosing LQTS or a QTc interval ≥480 ms offering 36% sensitivity and 100% specificity for diagnosing LQTS.[8] The latter cutoff has been incorporated into the updated Schwartz-score for the diagnosis of LQTS,[70] although it should be noted that this measure is only validated for patients with LQT1 and LQT2. Importantly, a prolonged QTc interval during exercise recovery is validated in children with LQTS, although the recommended duration of recovery monitoring is longer (optimal QTc interval measurement >460 ms at 7-minute recovery).[72] Changes in QTc interval during exercise and recovery may also be useful for distinguishing between LQTS genotypes, whereby patients with LQT1 have the greatest increase in QTc interval during exercise with gradual shortening during recovery, whereas patients with LQT2 have QTc interval shortening during exercise with gradual lengthening during recovery.[8,72,73] Consequently, differences in QTc interval during peak exercise or at 1-minute recovery could help differentiate between LQT1 and LQT2.

Abrupt standing from a supine position may also result in changes to the QTc interval, which can be performed in a clinic or incorporated into a standard exercise test (Supplemental Table 1).[74-78] Patients with LQTS exhibit both immediate and extended increases to their QTc interval when compared with control subjects (**Figure 5**, Supplemental Table 2). Overall, the measure with the highest diagnostic value during the stand-up test is the QTc interval during maximal QT stretch (defined as the time when the end of the T wave gets nearest to the next P-wave caused by R-R interval shortening without sufficient QT interval shortening),[76-78] with a

$QTc_{stretch}$ ≥490 ms offering 92% sensitivity and 79% specificity for diagnosing LQTS.[78] However, this measure is not validated in pediatric patients,[72,79] whereby abrupt standing can result in a QTc interval increase of up to 80 ms, even in healthy children.[80,81]

Changes in heart rate and repolarization are also described using pharmacologic provocation with epinephrine (**Figure 5**, Supplemental Table 1).[82-89] However, the heart rate response to epinephrine may be variable,[84] and induction of ventricular bigeminy or marked changes to T-wave morphology can adversely affect the test interpretation.[90,91] Therefore, the utility of the epinephrine challenge test as a stand-alone diagnostic tool for LQTS is limited, although it may be considered in patients who are unable to perform exercise.[92]

Active mental stress, such as mental arithmetic, may also provoke repolarization changes in patients with LQTS.[93,94] However, there is currently no universal agreement regarding the timing of QT measurements,[93-95] while the effects of baseline education and language proficiency may further confound results.[96] The active mental stress test requires standardization and clinical validation.

GENETIC TESTING. Genetic testing is recommended in those with a Schwartz-score ≥3,[3,37,97] although the identification of a culprit genetic variant ranges between 50% and 80% of cases depending on the pretest probability.[61,98,99] Additionally, differences in the global availability and cost of genetic testing also present as potential implementation barriers.[100] The 3 major genotypes—*KCNQ1* (~40%-45%), *KCNH2* (~40%) and *SCN5A* (~5%-10%) genes causing LQT1, LQT2, and LQT3, respectively[19,101]—account for the vast majority of genotype-positive LQTS cases and have the strongest level of evidence for causal pathogenicity.[3,22] Several "minor" LQTS genes (such as *CAV3, KCNE1*, *KCNE2*, *KCNJ2*, *KCNJ5,* and *SCN4B*) are now considered as "limited evidence" LQTS-susceptibility genes[3,22] resulting from the recent update of sequence variant interpretation as recommended by the American College of Medical Genetics.[102] The careful curation of identified gene variant(s) is critical before ascribing pathogenicity, and a practical approach would be to involve a genetic expert in this process. Interpretation should account for the frequency of the variant in population databases, its anticipated effect on protein function (based on in vitro and in vivo studies), and linkage studies. With the recent identification of possible polygenic influences in LQTS patients[103] and recognition regarding the need for periodic variant reinterpretation,[104] it is evident that our understanding of the genetics in LQTS is ever-evolving and therefore may be subject to further reclassification.[105]

FAMILY SCREENING. Cascade screening should be performed on all first-degree relatives of patients with LQTS. If a proband has an identified gene variant, targeted genetic testing should be performed in their first-degree relatives—family members who carry the gene variant should undergo clinical assessment. If a proband is gene elusive, all first-degree relatives should undergo clinical assessment. Screening for additional relatives (eg, second degree relatives) should be considered if an intervening relative cannot be tested, or in the presence of a clinical phenotype.

We would recommend that clinical assessment include baseline testing with ECG and exercise testing. Results from the initial evaluation would guide any additional testing and the frequency of follow-up testing.

OTHER TESTS. Although not necessary for the diagnosis, a baseline echocardiogram should be performed in all patients being assessed for LQTS for the exclusion of structural heart disease.

RISK STRATIFICATION

Patients with LQTS are at risk of developing serious arrhythmic events (SAE), encompassing SCD and resuscitated cardiac arrest. Thus, risk stratification in LQTS patients aims to identify those with a greater likelihood of SAE (**Table 3**). These are influenced by various clinical, ECG, and genetic factors, and an understanding of these factors can allow for shared decision-making regarding therapies.

CLINICAL. The risk of cardiac events (CE, which includes SAE and syncope) and SAE in LQTS appears greatest in childhood and gradually diminishes with time.[106-108] Older patients with LQTS—especially those aged >60 years—have an attenuated risk of SAE from LQTS compared with younger patients, and a higher competing risk of mortality from other conditions.[109]

The additional influence of patient sex results in an apparent age-sex interaction. In a cohort of 2,272 adolescent LQTS patients, Hobbs et al[110] found that male sex conferred a 4.0× relative risk of SAE between the ages of 10 and 12 years. This was reinforced by Costa et al[111] in a cohort of 1,051 LQT1 patients, where they found male sex was associated with a 2.3× relative risk of SAE up until age 13 years. During mid to late adolescence, there appears to be relatively

JACC: CLINICAL ELECTROPHYSIOLOGY VOL. 8, NO. 5, 2022
MAY 2022:687-706

TABLE 3 Summary of Select Risk Stratification Markers

Category First Author	Study Population	N	Outcome	Risk Marker	Comparator	Odds Ratio
Clinical						
Rashba[121]	LQT pregnancy	111	SAE	Postpartum	Pre-pregnancy	40.8
Hobbs[110]	LQT age 10-20 y	2,772	SAE	Male (10-12 y) Syncope β-Blocker	Female (10-12 y) No syncope No β-blocker	4.0 2.7-18.1 0.36
Sauer[112]	LQT ≥18 y	812	SAE	Female Syncope β-Blocker	Male No syncope No β-blocker	2.68 5.10 0.40
Moss[140]	LQT1	600	CE <40 y	Male (<13 y) β-blocker	Female (<13 y) No β-blocker	1.72-1.94 0.21-0.26
Seth[49]	LQT pregnancy	391	SAE	Postpartum	Pre-pregnancy	4.1
Goldenberg[109]	LQT, age 41-60 y	924	SAE	Recent syncope	No recent syncope	9.92
Jons[113]	LQT, QTc interval ≥450	1,059	SAE	Female (14-40 y) β-Blockers Syncope on β-blocker	Male (14-40 y) No β-blocker No syncope	1.86 0.46 3.59
Liu[122]	LQT, age <20 y	1,648	SAE	Syncope β-Blocker	No syncope No β-blocker	6.54-14.65 0.20-0.28
Migdalovich[123]	LQT2	1,166	SAE <40 y	Female (14-40 y) Syncope	Male (14-40 y) No syncope	2.23 3.15
Goldenberg[38]	LQT, QTc interval >440	1,392	SAE <40 y	Female (>13y)	Male (>13 y)	1.90
Costa[111]	LQT1	1,051	SAE <40 y	Male (0-13 y) Syncope	Female (0-13 y) No syncope	2.31 3.40
Mullally[114]	LQT	403	SAE <40 y	Female	Male	2.25
Laksman[115]	LQT	113	CE	Female	Male	5.21
Wilde[124]	LQT3	391	SAE <40 y	Syncope	No syncope	2.03
Koponen[130]	LQT1/2 age 18-40 y	867	CE	Female β-Blocker	Male No β-blocker	3.18 0.19-0.40
Sugrue[116]	LQT	651	CE	Female	Male	2.39
QTc interval						
Zareba[127]	LQT	246	CE <40 y	10-ms increment QTc interval		1.06
Priori[129]	LQT on β-blocker	335	CE	QTc interval >500	QTc interval ≤500	2.01
Goldenberg[125]	LQT age 10-20 y	375	CE	QTc interval ≥500	QTc interval <500	2.74
Sauer[112]	LQT ≥18 y	812	SAE	QTc interval ≥500	QTc interval <500	3.34-6.35
Moss[140]	LQT1	600	CE <40y	QTc interval ≥500	QTc interval <500	1.88-3.25
Jons[113]	LQT, QTc interval ≥450	1,059	SAE	QTc interval >500	QTc interval ≤500	1.76
Migdalovich[123]	LQT2	1,166	SAE <40 y	QTc interval ≥500	QTc interval <500	3.24
Goldenberg[38]	LQT, QTc interval >440	1,392	SAE <40 y	10-ms increment QTc interval		1.08
Costa[111]	LQT1	1,051	SAE <40 y	QTc interval ≥500	QTc interval <500	3.35-4.18
Mullally[114]	LQT	403	SAE <40 y	QTc interval ≥500	QTc interval <500	2.22
Laksman[115]	LQT	113	CE	QTc interval >500	QTc interval ≤500	4.50
Wilde[124]	LQT3	391	SAE <40 y	10-ms increment QTc interval		1.33
Koponen[130]	LQT1/2 age 18-40 y	867	CE	QTc interval ≥500	QTc interval <500	2.22-2.66
Shimizu[128]	LQT1/2	1,008	CE	10-ms increment QTc interval		1.06-1.09
Sugrue[116]	LQT	651	CE	10-ms increment QTc interval		1.07

Continued on the next page

similar risk between both sexes,[110,111] at least in those with LQT1.[110] Transitioning into adulthood, female sex results in an increased relative risk of both CE and SAE.[112-116] Sauer et al[112] showed that between ages 18 and 40 years, the rate of CE was 40% in females, compared with 14% in males, whereas the rate of SAE was 11% in females compared with 2% in males.

The reasons for these observations likely stem from the effects of sex hormones on potassium channels,[117] whereby testosterone and progesterone potentiate I_{Ks}, whereas estrogen inhibits I_{Kr}.[118] In postpubertal males, increased levels of testosterone result in a reduced QT interval.[119,120] In females, there appears to be competing effects of estrogen (which causes an increase in QT) and progesterone (which causes a decrease in QT),[119] although this requires further elucidation. For example, although both estrogen and progesterone increase during pregnancy and decrease during the postpartum period, relative differences in the hormones may explain why women experience more events during the postpartum phase with nonsignificant differences during pregnancy.[49,121] This effect is particularly evident in women with LQT2.[49]

TABLE 3 Continued

Category First Author	Study Population	N	Outcome	Risk Marker	Comparator	Odds Ratio
Genetic						
Zareba[127]	LQT	246	CE <40y	LQT1/2 LQT1	LQT3 LQT2	3.43-5.49 1.58
Moss[138]	LQT2	168	CE <40 y	Pore	Non-pore	10.7-11.7
Priori [36]	LQT	647	CE <40 y	LQT2/3	LQT1	1.61-1.80
Priori[129]	LQT on β-blocker	335	CE	LQT2/3	LQT1	2.81-4.00
Shimizu[139]	LQT1	66	CE	Transmembrane	C-terminus	3.4
Goldenberg[132]	LQT	2,218	CE	JLN syndrome	RW syndrome	5.51
Crotti[135]	LQT1	449	CE <40 y	A341V	Non-A341V	4.0
Moss[140]	LQT1	600	CE <40 y	Transmembrane Dominant negative	C-terminus Haploinsufficiency	2.06 2.26
Goldenberg[109]	LQT, age 41-60 y	539	SAE	LQT3	LQT1	4.53
Liu[136]	LQT3	85	CE <40 y	ΔKPQ	D1790G	2.42
Migdalovich[123]	LQT2 males	490	SAE <40 y	Pore-loop	Loop or C/N term	2.04-2.18
Goldenberg[38]	LQT, QTc interval ≤440 LQT, QTc interval >440	469 1,392	SAE <40 y	LQT1 Transmembrane LQT1	LQT2 Nontransmembrane LQT2	9.88 6.32 0.53
Wilde[124]	LQT3	391	SAE <40 y	E1784K/D1790 mutation		0.09-0.30
Koponen[130]	LQT1/2 age 18-40 y	867	CE	LQT2	LQT1	2.11
Mazzanti[131]	LQT	1,710	SAE	LQT2/3	LQT1	2.23-4.00
Shimizu[128]	LQT1/2	1,008	CE	S5-pore-S6	N/C terminus	1.64-1.99
Sugrue[116]	LQT	651	CE	LQT3	LQT1	0.28

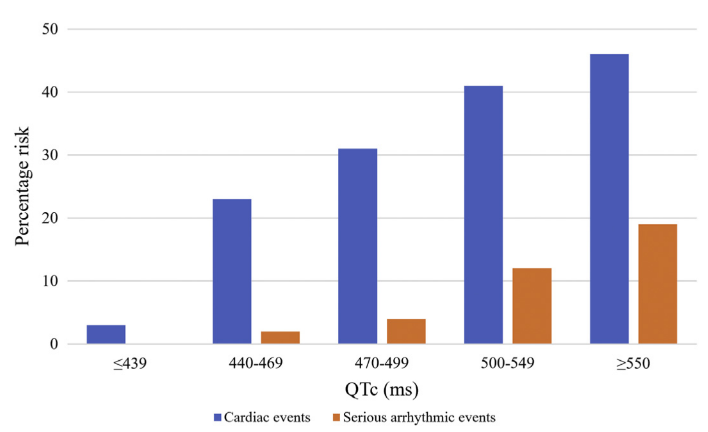
QTc values are in milliseconds.
CE = cardiac events; JLN = Jervell and Lange-Nielsen; LQT = long QT; LQTS = long QT syndrome; RW = Romano-Ward; SAE = serious arrhythmic events.

Finally, the occurrence of syncope is independently associated with the SAE with a relative risk of at least 2.0,[109-113,122-124] which is greater in those with recent (within 2 years) and/or recurrent syncope.[109,110,112,122]

QTc INTERVAL. QTc prolongation, which is the phenotypic manifestation of LQTS, imparts significant influence on the risk stratification of these patients. It should be noted that many of the risk-stratification studies are from the International LQTS Registry, which used the threshold method for QT measurement and included a cohort of patients with a more severe phenotype. Furthermore, it is important to consider the clinical utility of repeated measures of QTc interval, because studies have shown that the maximum measured QTc interval on serial ECG provides greater prediction of CE.[113,125,126]

Consistent evidence has shown that higher QTc values correlate with an increased incidence of CE (Figure 6).[36,38,112] Studies are typically retrospective cohorts with varying degrees of treatment (including many untreated subjects), so the cited risks combine natural history with on-treatment risk. In a study of 812 patients with genotype-positive LQTS (18% of cohort prescribed β-blockers), Sauer et al[112] showed a step-wise gradient of events. For patient groups with QTc interval ≤439 ms, 440 to 469 ms, 470 to 499 ms, 500 to 549 ms, and ≥550 ms, the corresponding risk of CE over 22 years (between ages 18 and 40 years) was 3%, 23%, 31%, 41%, and 46%, respectively, whereas the risk of SAE between ages 18 and 40 years was 0%, 2%, 4%, 12%, and 19%, respectively.[112]

FIGURE 6 Risk of Cardiac Events and Serious Arrhythmic Events Stratified by QTc Interval



Cardiac events = syncope + serious arrhythmic events. Serious arrhythmic events = resuscitated cardiac arrest + sudden cardiac death. Developed using original data reported by Sauer et al.[112] CE = cardiac events; SAE = serious arrhythmic events.

JACC: CLINICAL ELECTROPHYSIOLOGY VOL. 8, NO. 5, 2022
MAY 2022:687-706

This step-wise gradient is supported by findings of an increased HR for CE or SAE of ~1.1 for every 10-ms increment in QTc interval.[38,116,124,127,128] When dichotomized based on QTc interval, patients with a QTc interval ≥500 ms are ~2 to 3 times more likely to experience CE or SAE compared with LQTS patients with a QTc interval <500 ms,[110,111,114,123,125,129,130] including those who are medically managed with β-blockers.[129] Pragmatically, the application of a QTc ≥500 ms is a reasonable approach for the identification of high-risk patients with LQTS, because this corresponds with a 1.9% to 2.1%/year incidence of CE and 0.5% to 1.3%/year incidence of SAE.[112,131] By contrast, gene carriers with a QTc interval <500 ms have a low incidence of SAE at 0.2% to 0.4%/year,[112,131] whereas those with a QTc interval <460 ms are at very low risk for SAE at <0.1%/year.[38,112,131]

GENETIC. Although the autosomal recessive JLN form of LQTS and other compound heterozygote patients are associated with the most severe clinical outcomes,[107,132,133] additional genetic factors influence risk of both CE and SAE in LQTS.

With regard to genotype, initial work by Priori et al[36] found an annual incidence of SAE for LQT1 at 0.3%, LQT2 at 0.6%, and LQT3 at 0.6%. Our current understanding would estimate that the annual rate of SAE up to the age of 40 in LQT1 is 0.2% to 0.5%, LQT2 is 0.3% to 0.7%, and LQT3 is 0.3% to 1.1%.[111,112,123,124,131,134] Corresponding estimates for the annual rate of CE up to the age of 40 in LQT1 is 0.8% to 1.1%, LQT2 is 0.8% to 1.8%, and LQT3 is 0.5% to 1.5%.[36,55,112,124,135,136] Stratified further, Nannenberg et al[108] found that the specific period of increased mortality for LQT1 was between 1 and 19 years, for LQT2 was between 30 and 39 years, and for LQT3 was between 15 and 19 years.

With the advancement in the genetic understanding of LQTS, we now appreciate that there appears to be significant variation in the overall risk within long QT genotypes. In comparing a cohort of 438 genotype LQT1 patients (~50% prescribed β-blockers), Crotti et al[135] showed that presence of the A341V variant was associated with an 80% risk of experiencing a CE by age 40 years, compared with only 30% in those with LQT1 based on different causal variants. By contrast, Winbo et al[137] found a very low incidence (0.05% per year) of SAE in 80 Swedish patients (62.5% prescribed β-blockers) with the founder Y111C variant of LQT1 during 25 years of follow-up. Similarly, for cohorts of genotype-positive LQT3 patients, the presence of an E1784K or D1790G variant is associated with significantly less CE,[124,136] whereas those with ΔKPQ are at higher risk.[136]

Additional genetic markers have characterized variants depending on their location within the potassium channel. Consistent evidence indicates that variants affecting the transmembrane pore carry the highest relative risk of events, followed by variants affecting other transmembrane domains (but not specifically the pore region), with N or C terminus domains being associated with the lowest risk of events.[38,123,128,138-140] Furthermore, variants that result in a dominant negative effect convey greater risk than those resulting in haploinsufficiency.[140] Finally, various single nucleotide variation (formerly single nucleotide polymorphism) have been shown to influence the risk of both CE and SAE in LQTS.[126,141] These findings provide a plausible—at least partial—explanation for the variation in relative risk estimates when comparing subtypes of LQTS based on initial genetic studies.[36,127] Overall, the genetic risk stratification in LQTS is inherently complex and extends beyond the isolated identification of LQT subtype, requiring a multidisciplinary approach to variant interpretation and therapeutic recommendations.

OTHER FACTORS. Although LQTS is traditionally associated with normal left ventricular structure and function, cardiac imaging parameters may provide additional risk stratification.[116,142] In a cohort of 651 patients with LQTS, Sugrue et al[116] found that a more negative electromechanical window (defined as electromechanical window = time from QRS onset to aortic valve closure − QT interval for the same beat) was independently associated with the presence of symptoms. This interesting finding requires prospective evaluation and external validation.

MANAGEMENT

CONSERVATIVE. Nonpharmacologic management of LQTS involves the avoidance of QT-prolonging drugs and trigger avoidance. In those with asymptomatic LQTS and normal QTc interval (genotype positive, with little or no phenotype), conservative measures may be all that is required.[143]

An up-to-date list of drugs that can cause QT prolongation can be found at the CredibleMeds website or the related smartphone app,[144] including both prescription and nonprescription medications. Although this is an undoubtedly invaluable resource, it should be noted that the inclusion of most drugs is based on case report evidence with little systematic data regarding the degree of QT prolongation.[145] Nevertheless, it is crucial for patients to be educated about medications that are contraindicated in LQTS, especially given that there is suboptimal awareness

TABLE 4 Approach to Initiation of Possible QT-Prolonging Medication in LQTS Patient

1. Assessment of baseline risk (eg, age, sex, QTc interval) and protective factors (eg, ICD)
2. Consider inpatient admission in high-risk cases
3. If possible, select medication with least QT-prolonging effects from the appropriate class
4. Have an LQTS specialist available to interpret ECG if there is concern
5. Perform resting ECG before initiation
6. Use lowest possible dose for desired efficacy
7. Perform repeat ECG after reaching steady-state (typically 3-7 days) from medication initiation or dose increase
8. Contact LQTS specialist if QTc interval ≥500 ms or if there is a 30-ms increase in the QTc interval (would require additional surveillance with consideration of medication cessation)

Abbreviations as in Table 1.

among prescribers.[146] Patients should be cognizant that this also includes certain drugs commonly obtained as illicit substances such as amphetamine/methamphetamine, cocaine, oxycodone, and quetiapine. In cases where concomitant QT-prolonging medications are clearly indicated, consultation with a LQTS expert for shared decision-making and enhanced monitoring is advised (Table 4).

Our understanding regarding recreational and competitive sports in patients with LQTS is evolving. Based on cohort studies of predominantly pediatric and adolescent patients undertaking both recreational and competitive sports, the overall sport-related event rates in treated LQTS appears low.[147-150] Thus, in patients who are adequately assessed, treated and educated, sports participation can be considered appropriate.[151,152] The authors encourage recreational physical activity in medically treated, stable patients with its many health benefits. A shared decision-making process involving patients, families, and potentially sports team representatives to create a participation plan is strongly advised. One specific contraindication is swimming alone for LQT1 patients,[151] where the potential consequences of exertional syncope (which in itself may not be fatal) is compounded by the additional drowning risk.

PHARMACOLOGICAL. Pharmacological therapy with β-blockers is fundamental in the management of LQTS patients who are symptomatic for CE and/or have QTc prolongation. Treatment with β-blockers results in a minor reduction of resting QTc interval[153,154] and may ameliorate exercise-induced changes in QTc interval.[155,156] Pathophysiologically, β-blockers are thought to counteract the sympathetically induced heterogeneity in regional electrophysiology and reduces early after-depolarization triggers.[29,31] Large cohort studies consistently show that treatment with β-blockers results in a significant reduction of both CE and SAE (relative risk: 0.4-0.5 compared with those not treated with β-blockers).[110,112,113]

The treatment response is not universal across the β-blocker agents, and varies by genotype. The literature supporting β-blocker therapy pertains to its use in LQT1 and LQT2, with more data required with regard to LQT3.[157] Evidence suggests that treatment with nadolol results in the greatest risk reduction in the overall LQTS population,[131,158] being effective for both LQT1 and LQT2 patients.[158] Interestingly, propranolol may offer greater reductions in QTc interval,[153] although it is proposed that its efficacy is predominantly in patients with LQT1.[158] If nonselective agents cannot be tolerated, use of metoprolol, atenolol, or bisoprolol may be considered.[154,158]

Other important considerations in β-blocker therapy include adequate dosing, patient adherence, and its continued use during pregnancy. For nadolol, a target dose of 1 mg/kg/day is recommended,[159] with evening dosing or divided doses to minimize potential side effects. The occurrence of breakthrough CE and SAE are often described for patients who are nonadherent to β-blocker therapy.[148,160,161] In a study investigating pharmacy dispensing data in patients with LQTS, Waddell-Smith et al[162] found that over one-half of patients who are prescribed β-blockers had suboptimal adherence. This suggests that the relative risk reduction associated with optimal β-blocker therapy may be significantly greater than what is reported. In addition, continuation of β-blockers is considered safe during pregnancy[163,164] and reduces maternal risk especially in the postpartum period[49,121,165]; additional discussions about β-blocker use in LQTS and pregnancy is provided by Roston et al.[165] Thus, it is critical for clinicians to provide adequate education regarding the importance of β-blocker therapy for patients with LQTS.

A practical approach for assessing sufficient β-blocker effect is the demonstration of blunting of heart rate response with exercise testing. Rather than aiming for QTc shortening, the authors aim for heart rate blunting of 15% to 20% during maximum exercise to demonstrate adequate effect and adherence. Of note, certain patients may not require lifelong β-blocker therapy, such as those with shorter resting QTc intervals or older patients.[143]

Recently, the Class IB sodium channel-blocker mexiletine has demonstrated good efficacy as an adjunctive pharmacologic agent in the treatment of LQTS. In patients with LQT3, who have a gain-of-function variant in their $Na_V1.5$ sodium channel,

JACC: CLINICAL ELECTROPHYSIOLOGY VOL. 8, NO. 5, 2022
MAY 2022:687-706

mexiletine therapy at a dose of 8 mg/kg per day resulted in a significant (60 ms) reduction in QTc interval while reducing CE.[166] Interestingly, significant reductions in QTc interval have also been reported in patients with LQT2 receiving mexiletine.[167] The potential role for mexiletine in the treatment of LQTS warrants further investigation.

From small cohort studies, additional medical therapies that may shorten QTc interval include nicorandil (through its potentiation of potassium channels),[168] long-term potassium replacement therapy,[169] ranolazine (though sodium channel blockade) for LQT3,[170] and lumacaftor-ivacaftor (through potentiation of $K_V11.1$ channel trafficking).[171-173] These therapeutic options require validation in larger cohorts.

DEVICE THERAPY. In patients with LQTS and a history of resuscitated cardiac arrest, a secondary prevention implantable cardioverter-defibrillator (ICD) is indicated.[42] The decision regarding insertion of a primary prevention ICD is more challenging, requiring consideration of the absolute risk of SCD balanced against the absolute risk of device-related complications.[174,175] Two large retrospective cohort studies of patients with LQTS and ICD both found that a significantly prolonged QTc interval (≥500 ms) or cardiogenic syncope despite β-blocker therapy were independent predictors of appropriate ICD shocks,[176,177] whereas the presence of JLN syndrome is also known to portend to a significant SCD risk (35% by age 40 years despite β-blocker therapy).[132] By contrast, a meta-analysis of 462 LQTS patients with ICD found a 2.8% annual rate of inappropriate shocks, and a 7.0% annual rate of other complications such as lead malfunction, device infection, and psychological consequences.[178] Nevertheless, the authors would consider a shared decision-making discussion regarding primary prevention ICD (or left cardiac sympathetic denervation—see later in the text) for patients <40 years of age with JLN syndrome or compound heterozygotes (2 or more culprit variants), or patients who are adequately treated with β-blocker (± mexiletine) but still have a QTc interval ≥500 ms or recurrent cardiogenic syncope.

For patients undergoing ICD insertion, a dual-chamber transvenous device offers the potential benefit of atrial overdrive pacing during episodes of acute ventricular arrhythmias[179] or for the prevention of pause dependent QT prolongation. For this reason, there is currently minimal experience using subcutaneous ICD in patients with LQTS. From the limited available evidence in the EFFORTLESS S-ICD (Evaluation oF FactORs ImpacTing CLinical Outcome and Cost EffectiveneSS of the S-ICD) registry, patients with LQTS do not appear to be at increased risk of inappropriate shocks when compared with other patients.[180]

Pacemaker therapy (without ICD) may be considered in certain circumstances. In addition to preventing pause-dependent or bradycardia-related TdP,[181] pacing may ameliorate dynamic QT changes associated with heart rate variability. Long-term cardiac pacing may also result in a small reduction in overall QTc interval[182,183] and enable adequate titration of β-blocker therapy. In a small cohort of high-risk pediatric patients with JLN syndrome, the combination of atrial pacing and β-blockers was reported to be effective for the prevention of SAE.[184]

Finally, counselling for acquiring an automated external defibrillator may be considered in individuals with LQTS who do not undergo ICD insertion,[185] although the evidence for this course of action is limited.

LEFT CARDIAC SYMPATHETIC DENERVATION. Left cardiac sympathetic denervation (LCSD) can reduce the number of CE in patients who are already treated with β-blockers, or as potential monotherapy in those who are intolerant of β-blockers.[186-189] Despite the invasive nature and risk of complications, high-risk patients undergoing LCSD generally express satisfaction with the result.[190,191] LCSD is usually reserved for those who are intolerant of pharmacologic therapy or those with breakthrough CE after being treated with pharmacologic therapy. In certain high-risk patients (particularly LQT1), LCSD may be considered an appropriate or even preferred alternative intervention to the implantation of a primary prevention ICD.

FUTURE DIRECTIONS

Although patients with LQTS display a maladaptive repolarization response to changes in heart rate, there is still much to learn about the underlying pathophysiology. Recent studies have suggested that autonomic instability may contribute to symptoms and ventricular arrhythmias in patients with LQTS.[192,193] Furthermore, studies involving QT/QTc hysteresis in patients with LQTS indicate that their QT intervals are different during exercise when compared with recovery for a given heart rate,[57,75,77,194] suggesting repolarization variability depending on the level of sympathetic and parasympathetic input. Further work is required to delineate the role of the autonomic nervous system in relation to repolarization changes in patients with LQTS.

The diagnosis of LQTS is also evolving. For example, the foundations for the Schwartz-score was developed almost 30 years ago, before the identification of LQTS disease causing genes.[195] Although this is utilized by clinicians as a diagnostic tool, it is also a screening tool to determine which patients may be referred for genetic testing,[61] which in itself may lead to a diagnosis of LQTS. Additionally, phenotypic LQTS involves more than just the QT (and QTc) interval—as Peter Schwartz has reiterated on numerous occasions, "I don't measure the QT interval, I look at it."[4] Thus, the development of machine learning algorithms for the assessment of T-wave morphology,[196-198] and also genetic prediction,[199] may result in the improved detection of LQTS patients.[200]

Related to this is the potential for an algorithm-based risk stratification model. In addition to identifying T-wave morphology for diagnosis, the quantitative assessment of T-wave abnormalities may improve risk stratification.[201-203] Furthermore, there is currently an abundance of risk stratification markers that are independent predictors of CE and SAE when accounting for age, sex, QTc interval, and genotype. However, the interplay between these markers is largely unknown, whereas decisions based on risk stratification would benefit from establishing the annual SAE rate in addition to the relative risk.

The contemporary management of LQTS is evolving as clinicians become more exposed to those who have a genetic diagnosis with a more benign clinical phenotype.[204] Consequently, it would be important to identify patients in whom therapies are not necessarily required.[143] Finally, the refinement of medical therapies for LQTS is ongoing. For example, mexiletine significantly reduces the QTc interval in cohorts of LQT3 and LQT2 patients.[166,167] Whether this reduction in QTc interval leads to a reduction in CE and SAE in larger cohorts would be of interest. Of concern, mexiletine's recent designation as a neuromuscular "orphan drug" threatens the accessibility for patients with LQTS.[205] Other emerging therapies also show promise by providing a cellular ability to "correct" LQTS. Lumacaftor-ivacaftor aims to restore ion channel function,[171] and is currently used in patient with cystic fibrosis.[206] An in vivo study in 2 LQT2 patients, showed that lumacaftor-ivacaftor can result in significant QTc shortening, presumably by restoring potassium channel function.[172] Finally, in vitro studies using induced pluripotent stem cell-derived cardiomyocytes has indicated promise for *KCNQ1* ion channel antibodies for the restoration of I_{Ks} function,[207] *KCNQ1*-suppression-and-replacement gene therapy for the shortening of APD,[208], and serum and glucocorticoid kinase-1 inhibition that can ameliorate the gain-of-function of $Na_V1.5$ seen in LQT3.[209]

CONCLUSIONS

Congenital LQTS is a heterogenous group of conditions occurring caused by maladaptive cardiac repolarization. The assessment of clinical, ECG, and genetic factors are important for both the diagnostic evaluation and risk stratification of patients with LQTS. Management of patients with 1 of the 3 major LQTS genotypes requires an understanding of the various conservative, pharmacologic, and interventional treatment modalities, which are collectively very effective in mitigating risk.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

The study was supported by the Heart in Rhythm Organization (Dr Krahn, Principal Investigator), which receives support from the Canadian Institute of Health Research (RN380020-406814). Dr Krahn has received support from the Sauder Family and Heart and Stroke Foundation Chair in Cardiology (Vancouver, BC), the Paul Brunes Chair in Heart Rhythm Disorders (Vancouver, BC), and the Paul Albrechtson Foundation (Winnipeg, MB). Dr Laksman has received support from The University of British Columbia, Department of Medicine and the School of Biomedical Engineering, The University of British Columbia Cardiology Academic Practice Plan. Dr Ackerman receives support from The Mayo Clinic Windland Smith Rice Comprehensive Sudden Cardiac Death Program. Dr Wilde has received support from the Netherlands CardioVascular Research Initiative": the Dutch Heart Foundation, Dutch Federation of University Medical Centres, the Netherlands Organisation for Health Research and Development, and the Royal Netherlands Academy of Sciences (PREDICT2). Dr Han is supported by a Robert and Elizabeth Albert Travel Grant from the RACP Foundation, Australia. Dr Laksman is a consultant for Abbott, Medtronic, and Boston Scientific. Dr Ackerman is a consultant for Abbott, ARMGO Pharma, Boston Scientific, Daiichi-Sankyo, Invitae, LQT Therapeutics, Medtronic, and UpToDate; and holds equity in Alivecor and Anumana. Dr Wilde is a member of the scientific advisory board for LQT therapeutics. All other authors have reported that they have no relationships relevant to the contents of this paper to disclose.

ADDRESS FOR CORRESPONDENCE: Dr Andrew D. Krahn, Center for Cardiovascular Innovation, Heart Rhythm Services, Division of Cardiology, University of British Columbia, 211-1033 Davie Street, Vancouver, BC V6E 1M7, Canada. E-mail: akrahn@mail.ubc.ca. Twitter: @HeartsInRhythm.

REFERENCES

1. Moss AJ. Long QT syndrome. *JAMA*. 2003;289:2041–2044.

2. Goldenberg I, Moss AJ. Long QT syndrome. *J Am Coll Cardiol*. 2008;51:2291–2300.

3. Schwartz PJ, Crotti L, Insolia R. Long-QT syndrome: from genetics to management. *Circ Arrhythm Electrophysiol*. 2012;5:868–877.

4. Schwartz PJ, Ackerman MJ. The long QT syndrome: a transatlantic clinical approach to diagnosis and therapy. *Eur Heart J*. 2013;34:3109–3116.

5. Jervell A, Lange-Nielsen F. Congenital deaf-mutism, functional heart disease with prolongation of the Q-T interval and sudden death. *Am Heart J*. 1957;54:59–68.

6. Schwartz P, Periti M, Malliani A. Fundamentals of clinical cardiology: the long QT syndrome. *Am Heart J*. 1975;89:378–390.

7. Schwartz PJ. Idiopathic long QT syndrome: progress and questions. *Am Heart J*. 1985;109:399–411.

8. Sy RW, van der Werf C, Chattha IS, et al. Derivation and validation of a simple exercise-based algorithm for prediction of genetic testing in relatives of LQTS probands. *Circulation*. 2011;124:2187–2194.

9. Vink AS, Neumann B, Lieve KVV, et al. Determination and Interpretation of the QT Interval. *Circulation*. 2018;138:2345–2358.

10. Grant AO. Cardiac ion channels. *Circ Arrhythm Electrophysiol*. 2009;2:185–194.

11. Schmitt N, Grunnet M, Olesen SP. Cardiac potassium channel subtypes: new roles in repolarization and arrhythmia. *Physiol Rev*. 2014;94:609–653.

12. Tamargo J, Caballero R, Gómez R, Valenzuela C, Delpón E. Pharmacology of cardiac potassium channels. *Cardiovasc Res*. 2004;62:9–33.

13. Giudicessi JR, Ackerman MJ. Potassium-channel mutations and cardiac arrhythmias-diagnosis and therapy. *Nat Rev Cardiol*. 2012;9:319–332.

14. Nattel S, Carlsson L. Innovative approaches to anti-arrhythmic drug therapy. *Nat Rev Drug Discov*. 2006;5:1034–1049.

15. Lo CF, Numann R. Independent and exclusive modulation of cardiac delayed rectifying K+ current by protein kinase C and protein kinase A. *Circ Res*. 1998;83:995–1002.

16. Thomas D, Zhang W, Wu K, et al. Regulation of HERG potassium channel activation by protein kinase C independent of direct phosphorylation of the channel protein. *Cardiovasc Res*. 2003;59:14–26.

17. de Lera Ruiz M, Kraus RL. Voltage-gated sodium channels: structure, function, pharmacology, and clinical indications. *J Med Chem*. 2015;58:7093–7118.

18. Wilde AAM, Amin AS. Clinical spectrum of SCN5A mutations: long QT syndrome, Brugada syndrome, and cardiomyopathy. *J Am Coll Cardiol EP*. 2018;4:569–579.

19. Splawski I, Shen J, Timothy KW, et al. Spectrum of mutations in long-QT syndrome genes. KVLQT1, HERG, SCN5A, KCNE1, and KCNE2. *Circulation*. 2000;102:1178–1185.

20. Roberts JD, Krahn AD, Ackerman MJ, et al. Loss-of-function KCNE2 variants: true monogenic culprits of long-QT syndrome or proarrhythmic variants requiring secondary provocation? *Circ Arrhythm Electrophysiol*. 2017;10(8):e005282. https://doi.org/10.1161/CIRCEP.117.005282

21. Garmany R, Giudicessi JR, Ye D, Zhou W, Tester DJ, Ackerman MJ. Clinical and functional reappraisal of alleged type 5 long QT syndrome: causative genetic variants in the KCNE1-encoded minK β-subunit. *Heart Rhythm*. 2020;17:937–944.

22. Adler A, Novelli V, Amin AS, et al. An international, multicentered, evidence-based reappraisal of genes reported to cause congenital long QT syndrome. *Circulation*. 2020;141:418–428.

23. Schulze-Bahr E, Wang Q, Wedekind H, et al. KCNE1 mutations cause Jervell and Lange-Nielsen syndrome. *Nat Genet*. 1997;17:267–268.

24. Bennett PB, Yazawa K, Makita N, George AL Jr. Molecular mechanism for an inherited cardiac arrhythmia. *Nature*. 1995;376:683–685.

25. Moretti A, Bellin M, Welling A, et al. Patient-specific induced pluripotent stem-cell models for long-QT syndrome. *N Engl J Med*. 2010;363:1397–1409.

26. Itzhaki I, Maizels L, Huber I, et al. Modelling the long QT syndrome with induced pluripotent stem cells. *Nature*. 2011;471:225–229.

27. Malan D, Zhang M, Stallmeyer B, et al. Human iPS cell model of type 3 long QT syndrome recapitulates drug-based phenotype correction. *Basic Res Cardiol*. 2016;111:14.

28. Tyan L, Foell JD, Vincent KP, et al. Long QT syndrome caveolin-3 mutations differentially modulate K(v) 4 and Ca(v) 1.2 channels to contribute to action potential prolongation. *J Physiol*. 2019;597:1531–1551.

29. Saadeh K, Shivkumar K, Jeevaratnam K. Targeting the β-adrenergic receptor in the clinical management of congenital long QT syndrome. *Ann N Y Acad Sci*. 2020;1474:27–46.

30. Qu Z, Xie LH, Olcese R, et al. Early afterdepolarizations in cardiac myocytes: beyond reduced repolarization reserve. *Cardiovasc Res*. 2013;99:6–15.

31. Shimizu W, Ohe T, Kurita T, et al. Effects of verapamil and propranolol on early afterdepolarizations and ventricular arrhythmias induced by epinephrine in congenital long QT syndrome. *J Am Coll Cardiol*. 1995;26:1299–1309.

32. Schwartz PJ, Stramba-Badiale M, Crotti L, et al. Prevalence of the congenital long-QT syndrome. *Circulation*. 2009;120:1761–1767.

33. Vincent GM, Timothy KW, Leppert M, Keating M. The spectrum of symptoms and QT intervals in carriers of the gene for the long-QT syndrome. *N Engl J Med*. 1992;327:846–852.

34. Viskin S. The QT interval: too long, too short or just right. *Heart Rhythm*. 2009;6:711–715.

35. Straus SM, Kors JA, De Bruin ML, et al. Prolonged QTc interval and risk of sudden cardiac death in a population of older adults. *J Am Coll Cardiol*. 2006;47:362–367.

36. Priori SG, Schwartz PJ, Napolitano C, et al. Risk stratification in the long-QT syndrome. *N Engl J Med*. 2003;348:1866–1874.

37. Hofman N, Wilde AA, Kääb S, et al. Diagnostic criteria for congenital long QT syndrome in the era of molecular genetics: do we need a scoring system? *Eur Heart J*. 2007;28:575–580.

38. Goldenberg I, Horr S, Moss AJ, et al. Risk for life-threatening cardiac events in patients with genotype-confirmed long-QT syndrome and normal-range corrected QT intervals. *J Am Coll Cardiol*. 2011;57:51–59.

39. van der Werf C, Hofman N, Tan HL, et al. Diagnostic yield in sudden unexplained death and aborted cardiac arrest in the young: the experience of a tertiary referral center in The Netherlands. *Heart Rhythm*. 2010;7:1383–1389.

40. Krahn AD, Healey JS, Chauhan V, et al. Systematic assessment of patients with unexplained cardiac arrest: Cardiac Arrest Survivors With Preserved Ejection Fraction Registry (CASPER). *Circulation*. 2009;120:278–285.

41. Rucinski C, Winbo A, Marcondes L, et al. A population-based registry of patients with inherited cardiac conditions and resuscitated cardiac arrest. *J Am Coll Cardiol*. 2020;75:2698–2707.

42. Priori SG, Wilde AA, Horie M, et al. HRS/EHRA/APHRS expert consensus statement on the diagnosis and management of patients with inherited primary arrhythmia syndromes. *Heart Rhythm*. 2013;10:1932–1963.

43. Havakuk O, Viskin S. A tale of 2 diseases: the history of long-QT syndrome and Brugada syndrome. *J Am Coll Cardiol*. 2016;67:100–108.

44. Itoh H, Crotti L, Aiba T, et al. The genetics underlying acquired long QT syndrome: impact for genetic screening. *Eur Heart J*. 2016;37:1456–1464.

45. Viskin S, Alla SR, Barron HV, et al. Mode of onset of torsade de pointes in congenital long QT syndrome. *J Am Coll Cardiol*. 1996;28:1262–1268.

46. Wilde AA, Jongbloed RJ, Doevendans PA, et al. Auditory stimuli as a trigger for arrhythmic events differentiate HERG-related (LQTS2) patients from KVLQT1-related patients (LQTS1). *J Am Coll Cardiol*. 1999;33:327–332.

47. Moss AJ, Robinson JL, Gessman L, et al. Comparison of clinical and genetic variables of cardiac events associated with loud noise versus swimming among subjects with the long QT syndrome. *Am J Cardiol*. 1999;84:876–879.

48. Schwartz PJ, Priori SG, Spazzolini C, et al. Genotype-phenotype correlation in the long-QT syndrome: gene-specific triggers for life-threatening arrhythmias. *Circulation*. 2001;103:89–95.

49. Seth R, Moss AJ, McNitt S, et al. Long QT syndrome and pregnancy. *J Am Coll Cardiol*. 2007;49:1092–1098.

50. Toft E, Aarøe J, Jensen BT, et al. Long QT syndrome patients may faint due to neurocardiogenic syncope. *Europace*. 2003;5:367–370.

51. Johnson JN, Hofman N, Haglund CM, Cascino GD, Wilde AA, Ackerman MJ. Identification of a possible pathogenic link between congenital long QT syndrome and epilepsy. *Neurology*. 2009;72:24–231.

52. Auerbach DS, McNitt S, Gross RA, Zareba W, Dirksen RT, Moss AJ. Genetic biomarkers for the risk of seizures in long QT syndrome. *Neurology*. 2016;87:1660–1668.

53. Miyazaki A, Sakaguchi H, Aiba T, et al. Comorbid epilepsy and developmental disorders in congenital long QT syndrome with life-threatening perinatal arrhythmias. *J Am Coll Cardiol EP*. 2016;2:266–276.

54. Kirchhof P, Eckardt L, Franz MR, et al. Prolonged atrial action potential durations and polymorphic atrial tachyarrhythmias in patients with long QT syndrome. *J Cardiovasc Electrophysiol*. 2003;14:1027–1033.

55. Platonov PG, McNitt S, Polonsky B, et al. Risk stratification of type 2 long-QT syndrome mutation carriers with normal QTc interval: the value of sex, T-Wave morphology, and mutation type. *Circ Arrhythm Electrophysiol*. 2018;11:e005918.

56. Earle N, Crawford J, Smith W, et al. Community detection of long QT syndrome with a clinical registry: an alternative to ECG screening programs? *Heart Rhythm*. 2013;10:233–238.

57. Chattha IS, Sy RW, Yee R, et al. Utility of the recovery electrocardiogram after exercise: a novel indicator for the diagnosis and genotyping of long QT syndrome? *Heart Rhythm*. 2010;7:906–911.

58. Moss AJ, Schwartz PJ, Crampton RS, Locati E, Carleen E. The long QT syndrome: a prospective international study. *Circulation*. 1985;71:17–21.

59. Mönnig G, Eckardt L, Wedekind H, et al. Electrocardiographic risk stratification in families with congenital long QT syndrome. *Eur Heart J*. 2006;27:2074–2080.

60. Macfarlane PW, McLaughlin SC, Rodger JC. Influence of lead selection and population on automated measurement of QT dispersion. *Circulation*. 1998;98:2160–2167.

61. Taggart NW, Haglund CM, Tester DJ, Ackerman MJ. Diagnostic miscues in congenital long-QT syndrome. *Circulation*. 2007;115:2613–2620.

62. Bazett HC. An analysis of the time relations of electrocardiograms. *Heart*. 1920;7:353–370.

63. Vandenberk B, Vandael E, Robyns T, et al. Which QT correction formulae to use for QT monitoring? *J Am Heart Assoc*. 2016;5(6):e003264. https://doi.org/10.1161/JAHA.116.003264

64. Laksman ZW, Krahn AD. Fast and spurious: correcting the QT interval. *Heart Rhythm*. 2016;13:536–537.

65. Malfatto G, Beria G, Sala S, Bonazzi O, Schwartz PJ. Quantitative analysis of T wave abnormalities and their prognostic implications in the idiopathic long QT syndrome. *J Am Coll Cardiol*. 1994;23:296–301.

66. Zareba W, Moss AJ, le Cessie S, Hall WJ. T wave alternans in idiopathic long QT syndrome. *J Am Coll Cardiol*. 1994;23:1541–1546.

67. Moss AJ, Zareba W, Benhorin J, et al. ECG T-wave patterns in genetically distinct forms of the hereditary long QT syndrome. *Circulation*. 1995;92:2929–2934.

68. Zhang L, Timothy KW, Vincent GM, et al. Spectrum of ST-T-wave patterns and repolarization parameters in congenital long-QT syndrome: ECG findings identify genotypes. *Circulation*. 2000;102:2849–2855.

69. Kirchhof P, Franz MR, Bardai A, Wilde AM. Giant T-U waves precede torsades de pointes in long QT syndrome: a systematic electrocardiographic analysis in patients with acquired and congenital QT prolongation. *J Am Coll Cardiol*. 2009;54:143–149.

70. Schwartz PJ, Crotti L. QTc behavior during exercise and genetic testing for the long-QT syndrome. *Circulation*. 2011;124:2181–2184.

71. Vincent GM, Jaiswal D, Timothy KW. Effects of exercise on heart rate, QT, QTc and QT/QS2 in the Romano-Ward inherited long QT syndrome. *Am J Cardiol*. 1991;68:498–503.

72. Aziz PF, Wieand TS, Ganley J, et al. Genotype- and mutation site-specific QT adaptation during exercise, recovery, and postural changes in children with long-QT syndrome. *Circ Arrhythm Electrophysiol*. 2011;4:867–873.

73. Horner JM, Horner MM, Ackerman MJ. The diagnostic utility of recovery phase QTc during treadmill exercise stress testing in the evaluation of long QT syndrome. *Heart Rhythm*. 2011;8:1698–1704.

74. Walker BD, Krahn AD, Klein GJ, et al. Effect of change in posture and exercise on repolarization in patients with long QT syndrome with HERG channel mutations. *Can J Cardiol*. 2005;21:33–38.

75. Wong JA, Gula LJ, Klein GJ, Yee R, Skanes AC, Krahn AD. Utility of treadmill testing in identification and genotype prediction in long-QT syndrome. *Circ Arrhythm Electrophysiol*. 2010;3:120–125.

76. Viskin S, Postema PG, Bhuiyan ZA, et al. The response of the QT interval to the brief tachycardia provoked by standing: a bedside test for diagnosing long QT syndrome. *J Am Coll Cardiol*. 2010;55:1955–1961.

77. Adler A, van der Werf C, Postema PG, et al. The phenomenon of "QT stunning": the abnormal QT prolongation provoked by standing persists even as the heart rate returns to normal in patients with long QT syndrome. *Heart Rhythm*. 2012;9:901–908.

78. Chorin E, Havakuk O, Adler A, et al. Diagnostic value of T-wave morphology changes during "QT stretching" in patients with long QT syndrome. *Heart Rhythm*. 2015;12:2263–2271.

79. Vink AS, Hermans BJM, Pimenta J, et al. Diagnostic accuracy of the response to the brief tachycardia provoked by standing in children suspected for long QT syndrome. *Heart Rhythm. O2*. 2021;2:149–159.

80. Filippini L, Postema PG, Zoubin K, et al. The brisk-standing-test for long QT syndrome in prepubertal school children: defining normal. *Europace*. 2018;20:f108–f112.

81. Dionne A, Fournier A, Dahdah N, Abrams D, Khairy P, Abadir S. Dynamic QT interval changes from supine to standing in healthy children. *Can J Cardiol*. 2018;34:66–72.

82. Sun ZH, Swan H, Viitasalo M, Toivonen L. Effects of epinephrine and phenylephrine on QT interval dispersion in congenital long QT syndrome. *J Am Coll Cardiol*. 1998;31:1400–1405.

83. Tanabe Y, Inagaki M, Kurita T, et al. Sympathetic stimulation produces a greater increase in both transmural and spatial dispersion of repolarization in LQT1 than LQT2 forms of congenital long QT syndrome. *J Am Coll Cardiol*. 2001;37:911–919.

84. Ackerman MJ, Khositseth A, Tester DJ, Hejlik JB, Shen WK, Porter CB. Epinephrine-induced QT interval prolongation: a gene-specific paradoxical response in congenital long QT syndrome. *Mayo Clin Proc*. 2002;77:413–421.

85. Noda T, Takaki H, Kurita T, et al. Gene-specific response of dynamic ventricular repolarization to sympathetic stimulation in LQT1, LQT2 and LQT3 forms of congenital long QT syndrome. *Eur Heart J*. 2002;23:975–983.

86. Shimizu W, Noda T, Takaki H, et al. Epinephrine unmasks latent mutation carriers with LQT1 form of congenital long-QT syndrome. *J Am Coll Cardiol*. 2003;41:633–642.

87. Kaufman ES, Gorodeski EZ, Dettmer MM, Dikshteyn M. Use of autonomic maneuvers to probe phenotype/genotype discordance in congenital long QT syndrome. *Am J Cardiol*. 2005;96:1425–1430.

88. Vyas H, Hejlik J, Ackerman MJ. Epinephrine QT stress testing in the evaluation of congenital long-QT syndrome: diagnostic accuracy of the paradoxical QT response. *Circulation*. 2006;113:1385–1392.

89. Satomi K, Shimizu W, Takaki H, et al. Response of beat-by-beat QT variability to sympathetic stimulation in the LQT1 form of congenital long QT syndrome. *Heart Rhythm*. 2005;2:149–154.

90. Magnano AR, Talathoti N, Hallur R, Bloomfield DM, Garan H. Sympathomimetic infusion and cardiac repolarization: the normative effects of epinephrine and isoproterenol in healthy subjects. *J Cardiovasc Electrophysiol*. 2006;17:983–989.

91. Churet M, Luttoo K, Hocini M, Haïssaguerre M, Sacher F, Duchateau J. Diagnostic reproducibility of epinephrine drug challenge interpretation in suspected long QT syndrome. *J Cardiovasc Electrophysiol*. 2019;30:896–901.

92. Krahn AD, Healey JS, Chauhan VS, et al. Epinephrine infusion in the evaluation of unexplained cardiac arrest and familial sudden death: from the cardiac arrest survivors with preserved Ejection Fraction Registry. *Circ Arrhythm Electrophysiol*. 2012;5:933–940.

93. Haapalahti P, Viitasalo M, Perhonen M, et al. Comparison of QT peak and QT end interval responses to autonomic adaptation in asymptomatic LQT1 mutation carriers. *Clin Physiol Funct Imaging*. 2011;31:209–214.

94. Etienne P, Huchet F, Gaborit N, et al. Mental stress test: a rapid, simple, and efficient test to unmask long QT syndrome. *Europace*. 2018;20:2014–2020.

95. Paavonen KJ, Swan H, Piippo K, et al. Response of the QT interval to mental and physical stress in types LQT1 and LQT2 of the long QT syndrome. *Heart*. 2001;86:39–44.

96. Ardila A, Ostrosky-Solis F, Rosselli M, Gómez C. Age-related cognitive decline during normal aging: the complex effect of education. *Arch Clin Neuropsychol*. 2000;15:495–513.

97. Napolitano C, Priori SG, Schwartz PJ, et al. Genetic testing in the long QT syndrome: development and validation of an efficient approach to genotyping in clinical practice. *JAMA*. 2005;294:2975–2980.

98. Tester DJ, Will ML, Haglund CM, Ackerman MJ. Effect of clinical phenotype on yield of long QT syndrome genetic testing. *J Am Coll Cardiol*. 2006;47:764–768.

99. Hofman N, Tan HL, Alders M, et al. Yield of molecular and clinical testing for arrhythmia syndromes: report of 15 years' experience. *Circulation*. 2013;128:1513–1521.

100. Ormond KE, Laurino MY, Barlow-Stewart K, et al. Genetic counseling globally: Where are we now? *Am J Med Genet C Semin Med Genet*. 2018;178:98–107.

101. Tester DJ, Will ML, Haglund CM, Ackerman MJ. Compendium of cardiac channel mutations in 541 consecutive unrelated patients referred for long QT syndrome genetic testing. *Heart Rhythm*. 2005;2:507–517.

102. Richards S, Aziz N, Bale S, et al. Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. *Genet Med*. 2015;17:405–424.

103. Lahrouchi N, Tadros R, Crotti L, et al. Transethnic genome-wide association study provides insights in the genetic architecture and heritability of long QT syndrome. *Circulation*. 2020;142:324–338.

104. Davies B, Bartels K, Hathaway J, et al. Variant reinterpretation in survivors of cardiac arrest with preserved ejection fraction (the Cardiac Arrest Survivors With Preserved Ejection Fraction Registry) by clinicians and clinical commercial laboratories. *Circ Genom Precis Med*. 2021;14:e003235.

105. Giudicessi JR, Roden DM, Wilde AAM, Ackerman MJ. Classification and reporting of potentially proarrhythmic common genetic variation in long QT syndrome genetic testing. *Circulation*. 2018;137:619–630.

106. Brink PA, Crotti L, Corfield V, et al. Phenotypic variability and unusual clinical severity of congenital long-QT syndrome in a founder population. *Circulation*. 2005;112:2602–2610.

107. Schwartz PJ, Spazzolini C, Crotti L, et al. The Jervell and Lange-Nielsen syndrome: natural history, molecular basis, and clinical outcome. *Circulation*. 2006;113:783–790.

108. Nannenberg EA, Sijbrands EJ, Dijksman LM, et al. Mortality of inherited arrhythmia syndromes: insight into their natural history. *Circ Cardiovasc Genet*. 2012;5:183–189.

109. Goldenberg I, Moss AJ, Bradley J, et al. Long-QT syndrome after age 40. *Circulation*. 2008;117:2192–2201.

110. Hobbs JB, Peterson DR, Moss AJ, et al. Risk of aborted cardiac arrest or sudden cardiac death during adolescence in the long-QT syndrome. *JAMA*. 2006;296:1249–1254.

111. Costa J, Lopes CM, Barsheshet A, et al. Combined assessment of sex- and mutation-specific information for risk stratification in type 1 long QT syndrome. *Heart Rhythm*. 2012;9:892–898.

112. Sauer AJ, Moss AJ, McNitt S, et al. Long QT syndrome in adults. *J Am Coll Cardiol*. 2007;49:329–337.

113. Jons C, Moss AJ, Goldenberg I, et al. Risk of fatal arrhythmic events in long QT syndrome patients after syncope. *J Am Coll Cardiol*. 2010;55:783–788.

114. Mullally J, Goldenberg I, Moss AJ, et al. Risk of life-threatening cardiac events among patients with long QT syndrome and multiple mutations. *Heart Rhythm*. 2013;10:378–382.

115. Laksman ZW, Gula LJ, Saklani P, et al. Early repolarization is associated with symptoms in patients with type 1 and type 2 long QT syndrome. *Heart Rhythm*. 2014;11:1632–1638.

116. Sugrue A, van Zyl M, Enger N, et al. Echocardiography-guided risk stratification for long QT syndrome. *J Am Coll Cardiol*. 2020;76:2834–2843.

117. Asatryan B, Yee L, Ben-Haim Y, et al. Sex-related differences in cardiac channelopathies: implications for clinical practice. *Circulation*. 2021;143:739–752.

118. Odening KE, Koren G. How do sex hormones modify arrhythmogenesis in long QT syndrome? Sex hormone effects on arrhythmogenic substrate and triggered activity. *Heart Rhythm*. 2014;11:2107–2115.

119. Sedlak T, Shufelt C, Iribarren C, Merz CN. Sex hormones and the QT interval: a review. *J Womens Health (Larchmt)*. 2012;21(9):933–941. https://doi.org/10.1089/jwh.2011.3444

120. Vink AS, Clur SB, Geskus RB, et al. Effect of age and sex on the QTc interval in children and adolescents with type 1 and 2 long-QT syndrome. *Circ Arrhythm Electrophysiol*. 2017;10(4):e004645. https://doi.org/10.1161/CIRCEP.116.004645

121. Rashba EJ, Zareba W, Moss AJ, et al. LQTS Investigators. Influence of pregnancy on the risk for cardiac events in patients with hereditary long QT syndrome. *Circulation*. 1998;97:451–456.

122. Liu JF, Jons C, Moss AJ, et al. Risk factors for recurrent syncope and subsequent fatal or near-fatal events in children and adolescents with long QT syndrome. *J Am Coll Cardiol*. 2011;57:941–950.

123. Migdalovich D, Moss AJ, Lopes CM, et al. Mutation and gender-specific risk in type 2 long QT syndrome: implications for risk stratification for life-threatening cardiac events in patients with long QT syndrome. *Heart Rhythm*. 2011;8:1537–1543.

124. Wilde AA, Moss AJ, Kaufman ES, et al. Clinical aspects of type 3 long-QT syndrome: an international multicenter study. *Circulation*. 2016;134:872–882.

125. Goldenberg I, Mathew J, Moss AJ, et al. Corrected QT variability in serial electrocardiograms in long QT syndrome: the importance of the maximum corrected QT for risk stratification. *J Am Coll Cardiol*. 2006;48:1047–1052.

126. Tomás M, Napolitano C, De Giuli L, et al. Polymorphisms in the NOS1AP gene modulate QT interval duration and risk of arrhythmias in the long QT syndrome. *J Am Coll Cardiol*. 2010;55:2745–2752.

127. Zareba W, Moss AJ, Schwartz PJ, et al. International Long-QT Syndrome Registry Research Group. Influence of the genotype on the clinical course of the long-QT syndrome. *N Engl J Med*. 1998;339:960–965.

128. Shimizu W, Makimoto H, Yamagata K, et al. Association of genetic and clinical aspects of congenital long QT syndrome with life-threatening arrhythmias in Japanese patients. *JAMA Cardiol*. 2019;4:246–254.

129. Priori SG, Napolitano C, Schwartz PJ, et al. Association of long QT syndrome loci and cardiac events among patients treated with beta-blockers. *JAMA*. 2004;292:1341–1344.

130. Koponen M, Havulinna AS, Marjamaa A, et al. Clinical and molecular genetic risk determinants in adult long QT syndrome type 1 and 2 patients. *BMC Med Gen*. 2018;19(1):56. https://doi.org/10.1186/s12881-018-0574-0

131. Mazzanti A, Maragna R, Vacanti G, et al. Interplay between genetic substrate, QTc duration, and arrhythmia risk in patients with long QT syndrome. *J Am Coll Cardiol*. 2018;71:1663–1671.

132. Goldenberg I, Moss AJ, Zareba W, et al. Clinical course and risk stratification of patients affected with the Jervell and Lange-Nielsen syndrome. *J Cardiovasc Electrophysiol*. 2006;17:1161–1168.

133. Westenskow P, Splawski I, Timothy KW, Keating MT, Sanguinetti MC. Compound mutations: a common cause of severe long-QT syndrome. *Circulation*. 2004;109:1834–1841.

134. Barsheshet A, Peterson DR, Moss AJ, et al. Genotype-specific QT correction for heart rate and the risk of life-threatening cardiac events in adolescents with congenital long-QT syndrome. *Heart Rhythm*. 2011;8:1207–1213.

135. Crotti L, Spazzolini C, Schwartz PJ, et al. The common long-QT syndrome mutation KCNQ1/A341V causes unusually severe clinical manifestations in patients with different ethnic backgrounds: toward a mutation-specific risk stratification. *Circulation*. 2007;116:2366–2375.

136. Liu JF, Moss AJ, Jons C, et al. Mutation-specific risk in two genetic forms of type 3 long QT syndrome. *Am J Cardiol*. 2010;105:210–213.

137. Winbo A, Diamant UB, Stattin EL, Jensen SM, Rydberg A. Low incidence of sudden cardiac death in a Swedish Y111C type 1 long-QT syndrome population. *Circ Cardiovasc Genet*. 2009;2:558–564.

138. Moss AJ, Zareba W, Kaufman ES, et al. Increased risk of arrhythmic events in long-QT syndrome with mutations in the pore region of the human ether-a-go-go-related gene potassium channel. *Circulation*. 2002;105:794–799.

139. Shimizu W, Horie M, Ohno S, et al. Mutation site-specific differences in arrhythmic risk and sensitivity to sympathetic stimulation in the LQT1 form of congenital long QT syndrome: multicenter study in Japan. *J Am Coll Cardiol*. 2004;44:117–125.

140. Moss AJ, Shimizu W, Wilde AA, et al. Clinical aspects of type-1 long-QT syndrome by location, coding type, and biophysical function of mutations involving the KCNQ1 gene. *Circulation*. 2007;115: 2481–2489.

141. Earle N, Yeo Han D, Pilbrow A, et al. Single nucleotide polymorphisms in arrhythmia genes modify the risk of cardiac events and sudden death in long QT syndrome. *Heart Rhythm*. 2014;11:76–82.

142. ter Bekke RM, Haugaa KH, van den Wijngaard A, et al. Electromechanical window negativity in genotyped long-QT syndrome patients: relation to arrhythmia risk. *Eur Heart J*. 2015;36:79–186.

143. MacIntyre CJ, Rohatgi RK, Sugrue AM, Bos JM, Ackerman MJ. Intentional nontherapy in long QT syndrome. *Heart Rhythm*. 2020;17:1147–1150.

144. Woosley R, Heise C, Gallo T, Tate J, Woosley D, Romero K. QTdrugs List. Accessed February 8, 2022. https://www.crediblemeds.org/

145. Beach SR, Celano CM, Sugrue AM, et al. QT prolongation, torsades de pointes, and psychotropic medications: a 5-year update. *Psychosomatics*. 2018;59:105–122.

146. Weeke PE, Kellemann JS, Jespersen CB, et al. Long-term proarrhythmic pharmacotherapy among patients with congenital long QT syndrome and risk of arrhythmia and mortality. *Eur Heart J*. 2019;40:3110–3117.

147. Johnson JN, Ackerman MJ. Competitive sports participation in athletes with congenital long QT syndrome. *JAMA*. 2012;308:764–765.

148. Johnson JN, Ackerman MJ. Return to play? Athletes with congenital long QT syndrome. *Brit J Sports Med*. 2013;47:28–33.

149. Aziz PF, Sweeten T, Vogel RL, et al. Sports participation in genotype positive children with long QT syndrome. *J Am Coll Cardiol EP*. 2015;1: 62–70.

150. Chambers KD, Beausejour Ladouceur V, Alexander ME, et al. Cardiac events during competitive, recreational, and daily activities in children and adolescents with long QT syndrome. *J Am Heart Assoc*. 2017;6(9):e005445. https://doi.org/10.1161/JAHA.116.005445

151. Ackerman MJ, Zipes DP, Kovacs RJ, Maron BJ. Eligibility and disqualification recommendations for competitive athletes with cardiovascular abnormalities: Task Force 10: the cardiac channelopathies: a scientific statement from the American Heart Association and American College of Cardiology. *J Am Coll Cardiol*. 2015;66:2424–2428.

152. Tobert KE, Bos JM, Garmany R, Ackerman MJ. Return-to-play for athletes with long QT syndrome or genetic heart diseases predisposing to sudden death. *J Am Coll Cardiol*. 2021;78(6): 594–604.

153. Chockalingam P, Crotti L, Girardengo G, et al. Not all beta-blockers are equal in the management of long QT syndrome types 1 and 2: higher recurrence of events under metoprolol. *J Am Coll Cardiol*. 2012;60:2092–2099.

154. Steinberg C, Padfield GJ, Al-Sabeq B, et al. Experience with bisoprolol in long-QT1 and long-QT2 syndrome. *J Interv Card Electrophysiol*. 2016;47:163–170.

155. Krahn AD, Yee R, Chauhan V, et al. Beta blockers normalize QT hysteresis in long QT syndrome. *Am Heart J*. 2002;143:528–534.

156. Bennett MT, Gula LJ, Klein GJ, et al. Effect of beta-blockers on QT dynamics in the long QT syndrome: measuring the benefit. *Europace*. 2014;16:1847–1851.

157. Ahn J, Kim HJ, Choi JI, et al. Effectiveness of beta-blockers depending on the genotype of congenital long-QT syndrome: a meta-analysis. *PLoS One*. 2017;12:e0185680.

158. Abu-Zeitone A, Peterson DR, Polonsky B, McNitt S, Moss AJ. Efficacy of different beta-blockers in the treatment of long QT syndrome. *J Am Coll Cardiol*. 2014;64:1352–1358.

159. Ackerman MJ, Priori SG, Dubin AM, et al. Beta-blocker therapy for long QT syndrome and catecholaminergic polymorphic ventricular tachycardia: are all beta-blockers equivalent? *Heart Rhythm*. 2017;14:e41–e44.

160. Dorostkar PC, Eldar M, Belhassen B, Scheinman MM. Long-term follow-up of patients with long-QT syndrome treated with beta-blockers and continuous pacing. *Circulation*. 1999;100:2431–2436.

161. Moss AJ, Zareba W, Hall WJ, et al. Effectiveness and limitations of beta-blocker therapy in congenital long-QT syndrome. *Circulation*. 2000;101:616–623.

162. Waddell-Smith KE, Li J, Smith W, Crawford J, Skinner JR. β-Blocker adherence in familial long QT syndrome. *Circ Arrhythm Electrophysiol*. 2016;9(8):e003591. https://doi.org/10.1161/CIRCEP.115.003591

163. Duan L, Ng A, Chen W, et al. β-Blocker exposure in pregnancy and risk of fetal cardiac anomalies. *JAMA Intern Med*. 2017;177:885–887.

164. Bateman BT, Heide-Jørgensen U, Einarsdóttir K, et al. β-Blocker use in pregnancy and the risk for congenital malformations: an international cohort study. *Ann Intern Med*. 2018;169:665–673.

165. Roston TM, van der Werf C, Cheung CC, et al. Caring for the pregnant woman with an inherited arrhythmia syndrome. *Heart Rhythm*. 2020;17: 341–348.

166. Mazzanti A, Maragna R, Faragli A, et al. Gene-specific therapy with mexiletine reduces arrhythmic events in patients with long QT syndrome type 3. *J Am Coll Cardiol*. 2016;67:1053–1058.

167. Bos JM, Crotti L, Rohatgi RK, et al. Mexiletine shortens the QT interval in patients with potassium channel-mediated type 2 long QT syndrome. *Circ Arrhythm Electrophysiol*. 2019;12: e007280.

168. Shimizu W, Kurita T, Matsuo K, et al. Improvement of repolarization abnormalities by a K+ channel opener in the LQT1 form of congenital long-QT syndrome. *Circulation*. 1998;97:1581–1588.

169. Etheridge SP, Compton SJ, Tristani-Firouzi M, Mason JW. A new oral therapy for long QT syndrome: long-term oral potassium improves repolarization in patients with HERG mutations. *J Am Coll Cardiol*. 2003;42:1777–1782.

170. Chorin E, Hu D, Antzelevitch C, et al. Ranolazine for congenital long-QT syndrome type III: experimental and long-term clinical data. *Circ Arrhythm Electrophysiol*. 2016;9(10):e004370. https://doi.org/10.1161/CIRCEP.116.004370

171. Mehta A, Ramachandra CJA, Singh P, et al. Identification of a targeted and testable antiarrhythmic therapy for long-QT syndrome type 2 using a patient-specific cellular model. *Eur Heart J*. 2018;39:1446–1455.

172. Schwartz PJ, Gnecchi M, Dagradi F, et al. From patient-specific induced pluripotent stem cells to clinical translation in long QT syndrome Type 2. *Eur Heart J*. 2019;40:1832–1836.

173. O'Hare BJ, John Kim CS, Hamrick SK, Ye D, Tester DJ, Ackerman MJ. Promise and potential peril with lumacaftor for the trafficking defective type 2 long-QT syndrome-causative variants, p. G604S, p.N633S, and p.R685P, using patient-specific re-engineered cardiomyocytes. *Circ Genom Precis Med*. 2020;13:466–475.

174. Etheridge SP, Sanatani S, Cohen MI, Albaro CA, Saarel EV, Bradley DJ. Long QT syndrome in children in the era of implantable defibrillators. *J Am Coll Cardiol*. 2007;50:1335–1340.

175. Gaba P, Bos JM, Cannon BC, et al. Implantable cardioverter-defibrillator explantation for overdiagnosed or overtreated congenital long QT syndrome. *Heart Rhythm*. 2016;13:879–885.

176. Schwartz PJ, Spazzolini C, Priori SG, et al. Who are the long-QT syndrome patients who receive an implantable cardioverter-defibrillator and what happens to them?: data from the European Long-QT Syndrome Implantable Cardioverter-Defibrillator (LQTS ICD) Registry. *Circulation*. 2010;122:1272–1282.

177. Biton Y, Rosero S, Moss AJ, et al. Primary prevention with the implantable cardioverter-defibrillator in high-risk long-QT syndrome patients. *Europace*. 2019;21:339–346.

178. Olde Nordkamp LR, Postema PG, Knops RE, et al. Implantable cardioverter-defibrillator harm in young patients with inherited arrhythmia syndromes: a systematic review and meta-analysis of inappropriate shocks and complications. *Heart Rhythm*. 2016;13:443–454.

179. Laksman Z, Barichello S, Roston TM, Deyell MW, Krahn AD. Acute management of ventricular arrhythmia in patients with suspected inherited heart rhythm disorders. *J Am Coll Cardiol EP*. 2019;5:267–283.

180. Lambiase PD, Eckardt L, Theuns DA, et al. Evaluation of subcutaneous implantable cardioverter-defibrillator performance in patients with ion channelopathies from the EFFORTLESS cohort and comparison with a meta-analysis of transvenous ICD outcomes. *Heart Rhythm O2*. 2020;1:326-335.

181. Viskin S. Cardiac pacing in the long QT syndrome: review of available data and practical recommendations. *J Cardiovasc Electrophysiol*. 2000;11:593-600.

182. Eldar M, Griffin JC, Van Hare GF, et al. Combined use of beta-adrenergic blocking agents and long-term cardiac pacing for patients with the long QT syndrome. *J Am Coll Cardiol*. 1992;20:830-837.

183. Kowlgi GN, Giudicessi JR, Barake W, Bos JM, Ackerman MJ. Efficacy of intentional permanent atrial pacing in the long-term management of congenital long QT syndrome. *J Cardiovasc Electrophysiol*. 2021;32:782-789.

184. Früh A, Siem G, Holmström H, Døhlen G, Haugaa KH. The Jervell and Lange-Nielsen syndrome; atrial pacing combined with ß-blocker therapy, a favorable approach in young high-risk patients with long QT syndrome? *Heart Rhythm*. 2016;13:2186-2192.

185. Pundi KN, Bos JM, Cannon BC, Ackerman MJ. Automated external defibrillator rescues among children with diagnosed and treated long QT syndrome. *Heart Rhythm*. 2015;12:776-781.

186. Schwartz PJ, Priori SG, Cerrone M, et al. Left cardiac sympathetic denervation in the management of high-risk patients affected by the long-QT syndrome. *Circulation*. 2004;109:1826-1833.

187. Bos JM, Bos KM, Johnson JN, Moir C, Ackerman MJ. Left cardiac sympathetic denervation in long QT syndrome: analysis of therapeutic nonresponders. *Circ Arrhythm Electrophysiol*. 2013;6:705-711.

188. Niaz T, Bos JM, Sorensen KB, Moir C, Ackerman MJ. Left cardiac sympathetic denervation monotherapy in patients with congenital long QT syndrome. *Circ Arrhythm Electrophysiol*. 2020;13:e008830.

189. Anderson HN, Bos JM, Rohatgi RK, Ackerman MJ. The effect of left cardiac sympathetic denervation on exercise in patients with long QT syndrome. *J Am Coll Cardiol EP*. 2019;5:1084-1090.

190. Waddell-Smith KE, Ertresvaag KN, Li J, et al. Physical and psychological consequences of left cardiac sympathetic denervation in long-QT syndrome and catecholaminergic polymorphic ventricular tachycardia. *Circ Arrhythm Electrophysiol*. 2015;8:1151-1158.

191. Antiel RM, Bos JM, Joyce DD, et al. Quality of life after videoscopic left cardiac sympathetic denervation in patients with potentially life-threatening cardiac channelopathies/cardiomyopathies. *Heart Rhythm*. 2016;13:62-69.

192. Porta A, Girardengo G, Bari V, et al. Autonomic control of heart rate and QT interval variability influences arrhythmic risk in long QT syndrome type 1. *J Am Coll Cardiol*. 2015;65:367-374.

193. Winter J, Tipton MJ, Shattock MJ. Autonomic conflict exacerbates long QT associated ventricular arrhythmias. *J Mol Cell Cardiol*. 2018;116:145-154.

194. Krahn AD, Klein GJ, Yee R. Hysteresis of the RT interval with exercise: a new marker for the long-QT syndrome? *Circulation*. 1997;96:1551-1556.

195. Schwartz PJ, Moss AJ, Vincent GM, Crampton RS. Diagnostic criteria for the long QT syndrome. An update. *Circulation*. 1993;88:782-784.

196. Badilini F, Vaglio M, Dubois R, et al. Automatic analysis of cardiac repolarization morphology using Gaussian mesa function modeling. *J Electrocardiol*. 2008;41:588-594.

197. Hermans BJM, Stoks J, Bennis FC, et al. Support vector machine-based assessment of the T-wave morphology improves long QT syndrome diagnosis. *Europace*. 2018;20:iii113-iii119.

198. Hermans BJM, Bennis FC, Vink AS, et al. Improving long QT syndrome diagnosis by a polynomial-based T-wave morphology characterization. *Heart Rhythm*. 2020;17:752-758.

199. Li B, Mendenhall JL, Kroncke BM, et al. Predicting the functional impact of KCNQ1 variants of unknown significance. *Circ Cardiovasc Genet*. 2017;10(5):e001754. https://doi.org/10.1161/CIRCGENETICS.117.001754

200. Giudicessi JR, Schram M, Bos JM, et al. Artificial intelligence-enabled assessment of the heart rate corrected QT interval using a mobile electrocardiogram device. *Circulation*. 2021;143(13):1274-1286.

201. Samol A, Gönes M, Zumhagen S, et al. Improved clinical risk stratification in patients with long QT Syndrome? Novel insights from multi-channel ECGs. *PLoS One*. 2016;11:e0158085.

202. Sugrue A, Rohatgi RK, Noseworthy PA, et al. Architectural T-wave analysis and identification of on-therapy breakthrough arrhythmic risk in type 1 and type 2 long-QT syndrome. *Circ Arrhythm Electrophysiol*. 2017;10(11):e005648. https://doi.org/10.1161/CIRCEP.117.005648

203. Cortez D, Zareba W, McNitt S, Polonsky B, Rosero SZ, Platonov PG. Quantitative T-wave morphology assessment from surface ECG is linked with cardiac events risk in genotype-positive KCNH2 mutation carriers with normal QTc values. *J Cardiovasc Electrophysiol*. 2019;30:2907-2913.

204. Rohatgi RK, Sugrue A, Bos JM, et al. Contemporary outcomes in patients with long QT syndrome. *J Am Coll Cardiol*. 2017;70:453-462.

205. Postema PG, Schwartz PJ, Arbelo E, et al. Continued misuse of orphan drug legislation: a life-threatening risk for mexiletine. *Eur Heart J*. 2020;41:614-617.

206. Wainwright CE, Elborn JS, Ramsey BW, et al. Lumacaftor-ivacaftor in patients with cystic fibrosis homozygous for Phe508del CFTR. *N Engl J Med*. 2015;373:220-231.

207. Maguy A, Kucera JP, Wepfer JP, Forest V, Charpentier F, Li J. KCNQ1 antibodies for immunotherapy of long QT syndrome type 2. *J Am Coll Cardiol*. 2020;75:2140-2152.

208. Dotzler SM, Kim CSJ, Gendron WAC, et al. Suppression-replacement KCNQ1 gene therapy for type 1 long QT syndrome. *Circulation*. 2021;143:1411-1425.

209. Bezzerides VJ, Zhang A, Xiao L, et al. Inhibition of serum and glucocorticoid regulated kinase-1 as novel therapy for cardiac arrhythmia disorders. *Sci Rep*. 2017;7:346.

KEY WORDS arrhythmia, inherited, sudden death

APPENDIX For supplemental tables, and references, please see the online version of this paper.